

Exploring Student Mental Health and Wellbeing in the Context of Campus Spaces

by

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Abstract

The built environment has implications for mental health and wellbeing, but there is little peer-reviewed research addressing this relationship, particularly in the context of post-secondary institutions. This is significant given the prevalence of mental health and wellbeing issues experienced by students.

This foundational study addresses two primary research questions: *1) What is the relationship between the post-secondary campus built environment and students' mental health and wellbeing?* and *2) Are coping strategies and design features on campus related to students' considerations of where to go when experiencing different emotions?*

My findings reveal the importance of student centres and libraries in the space decision-making of students. Additionally, I note that students rely more on avoidance-based and emotion-based coping strategies as motivation for space selection. Design features such as layout, cleanliness, natural light, and noise control may also influence students' space choices.

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List of Acronyms

CMHA	Canadian Mental Health Association
CSA	Canadian Standards Association
GRR	General resistance resources
ISO	International Standards Organizations or International Organization for Standardization
SRR	Specific resistance resources

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Chapter 1: Introduction

Over the course of any given day, people spend most of their time surrounded by constructed environments that can have significant implications for emotional, psychological, and physical health and wellbeing (Evans, 2003; Shepley & Pasha, 2017). While earlier research explores mental health and wellbeing on post-secondary campuses, there is still uncertainty in how post-secondary student experiences within the *built environment* contribute to their mental health. This gap seems significant given the prevalence of mental health and wellbeing issues experienced by students during the pursuit of post-secondary education (Mental Health Commission of Canada, 2020; *Mental Illness and Addiction: Facts and Statistics* | CAMH, n.d.). Furthermore, universities are somewhat unique as they have a fiduciary responsibility for the safety and duty of care of their students (Shanahan, 2019).

This foundational study addresses two broad research questions:

1. *Study 1: Within the literature, what relationship, if any, exists between the post-secondary campus built environment and students' mental health and wellbeing?*

To address this question, I conducted a rapid scoping review of peer-reviewed academic literature focused on the intersection of student mental health and the built environment on post-secondary campuses.

2. *Study 2: How might coping strategies and design features on campus influence students' considerations of where to go when experiencing different emotions?*

Using a questionnaire, I probed how students go to or avoid particular campus spaces when they felt certain emotions and why. The questionnaire explored three sub-questions:

- a. What campus spaces did students go to or avoid when experiencing certain emotions?
- b. What campus spaces were perceived to facilitate certain coping strategies?
- c. What was the reported influence of particular design features in campus spaces?

Results from both research questions help inform a preliminary or partial understanding of the context of a post-secondary campus space in relation to student mental health and wellbeing.

My thesis has five chapters. The first two chapters present the introduction to the study (Chapter 1) and background literature (Chapter 2). The methods, results and discussion for study one and two are presented in two separate chapters as they build off one another (Chapters 3 and 4), followed by the discussion (Chapter 5) and conclusion (Chapter 6).

1.1 Rationale

Three main thoughts motivated this research project. Firstly, in North America, we spend more than 90% of our time in the built environment (Evans, 2003a) and this raised my curiosity about how it may influence the way we may think, feel, and behave. Secondly, there is a mental health crisis among youth today, impacting youth between the ages they are most likely to attend post-secondary institutions (Linden et al., 2021; Mental Health - Ontario's Universities, n.d.; Mental Illness and Addiction: Facts and Statistics | CAMH, n.d.). Thirdly, post-secondary institutions are in an interesting position in that they both provide the built environment and have a responsibility for health, wellbeing, and safety of their students. After a brief literature search, there seems to be a gap in academic research bringing these three lines of thinking together. This initial search is fundamental in helping design the scope of study one, a rapid scoping review that

probes deeper into academic literature about postsecondary campus built environments and the mental health and wellbeing of students.

The results of study one informs study two, the latter of which examines how students report engaging with the built environment on a post-secondary campus and how they perceive their choices to be associated with their emotional states and coping needs. Design is an essential component of any built space, comprising physical elements, such as interior design, layout, and occupational health features (e.g., thermal comfort, lighting), all of which may influence psychosocial responses. This applied research looks to build on the existing knowledge base regarding the influence of the built environment on the mental health and wellbeing of post-secondary students while attempting to inform the development of future post-secondary campus initiatives to support this important aspect of health.

1.2 Definitions

The *built environment* refers to elements purposely designed and constructed (Healthy Built Environment | PlanH, n.d.). In this study, the built environment includes campus buildings (interiors and building envelop), landscape architecture and other deliberately designed areas of campus. *Mental health* refers to the overall state of one's mental wellbeing, which plays a crucial role in how we handle stress, regulate our emotion, and actively participate in our daily lives and community (Inwood & Ferrari, 2018; World Health Organization, 2022). Mental health is an essential aspect of our overall health as it forms the foundation for our ability to make informed decisions, foster meaningful relationships, and contribute to society (World Health Organization, 2022). *Wellbeing* (or well-being) refers to a general sense of one's mental state and acts as a

resource for daily life (CSA, 2020; World Health Organization, 2021). The study employs both terms to capture the continuum of health (i.e., mental health and wellbeing).

1.3 Contribution

Outputs from this research may provide insight into how emotions and coping strategies contribute to our understanding of how students experience the campus built environment, an area that appears to be sparsely studied in academic research. The insights gained from this project may inform different stakeholders that could benefit from the knowledge, including post-secondary units (e.g., facilities management, student mental health promotion units), strategic plan working groups, and standards and evaluation organizations, such as the Rick Hansen Foundation or the Canadian Standards Association.

Chapter 2: Background

2.1 The built environment and its relationship with student mental health

The built environment can facilitate mental health and wellbeing by supporting access to social, emotional-, spiritual-, avoidance- or problem-based coping strategies. For instance, after a difficult interaction with a peer, a student may experience the campus built environment by going to a prayer room (e.g., spiritual-based), accessing drop-in counselling (e.g., problem-based), meeting their friends at a campus café (e.g., social), going to the gym and letting off steam by working out (e.g., emotion-based), or going to their residence room and watch a movie (e.g., avoidance-based). Post-secondary institutions are novel environments to study the built environment's influence on mental health and wellbeing given the reliance of many students on the campus infrastructure as they engage in various activities (e.g., teaching and learning, research, housing, nutrition, exercise, and socialization). At the same time, the institution has a duty of care to deliver its services and perform its functions while protecting student health, safety, and welfare (Shanahan, 2019).

Looking at the built environment more broadly, many peer-reviewed studies report on how aspects of it impact mental wellbeing (Beemer et al., 2019; Evans, 2003; Hoisington et al., 2019). Evans (2003) posits that characteristics of the built environment affect people in two ways: directly (e.g., housing, crowding, noise, indoor air quality, and light) and indirectly (e.g., personal control, social support, and restoration) (Figure 1). Additionally, Codinhoto et al., (2019) and Sullivan & Chang (2011) emphasize that the built environment can result in chronic

stress, leading to psychological outcomes such as increased agitation, anxiety, depression, learned helplessness, and emotional exhaustion.

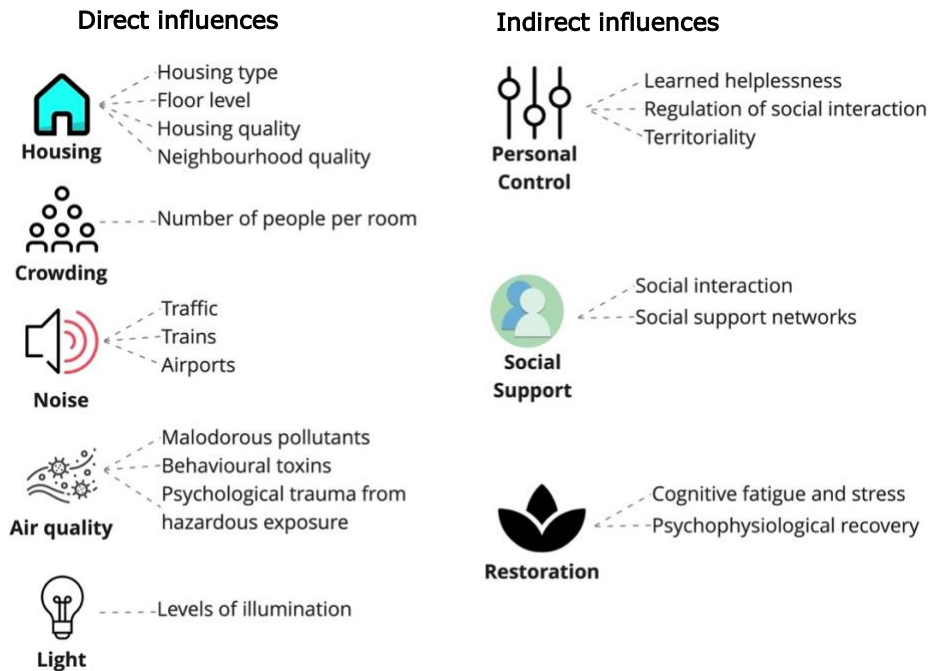


Figure 1 - Direct and indirect influences of the built environment (Evans, 2003)

Based on a literature review that encompassed research in urban settings, healthcare environments, workplaces, and standards there are specific design elements (Table 1) that could contribute to a state of mental wellbeing (Brennan et al., 2002; CSA, 2020; DuBose et al., 2018; Evans, 2003; Hoisington et al., 2019; Parsons, 2015; Shepley et al., 2016; Vischer & Wifi, 2015).

Table 1 - List of design features that could influence mental health and wellbeing

Design feature	Brennan, 2002	Evans, 2003	Parsons, 2015	Vischer, 2015	DuBose, MacAllister, et al., 2016	Shepley, Watson, et al., 2016	Hoisington, 2019	CSA Z2003: 20
Layout of the space	x	x	x	x	x			
Access to comfortable furniture			x	x		x		
Space is orderly and clean						x		
Quality of indoor lighting	x	x	x		x	x		
Access to natural lighting	x			x	x		x	
Access to a view of nature		x		x	x	x	x	
Noise control in the space	x	x	x	x	x	x	x	
Temperature control in the space	x		x	x			x	
Indoor air quality	x	x	x	x			x	
Space allows them to connect meaningfully with others		x				x		
Space fosters a sense of community		x						x
Space is flexible and accessible to adapt to diverse needs and activities								x
Space supports diverse cultural practices and ceremonies (i.e., smudging, ablution, prayer)								

Table 1 - List of design features that could influence mental health and wellbeing

Although Evans takes care to separate the direct and indirect influences, there is an interconnectedness among them. Collectively these factors contribute to the social determinants of health, social factors which play a critical role in promoting or undermining mental health and wellbeing (Raphael et al., 2020). This weaving of the physical and social spheres is reflected in the Canadian Standard Association CSA Z2003-20: Mental health and well-being for post-secondary students. Although the standard focuses primarily on social environments (e.g., the design and implementation of services, supports, and programs), it does highlight the need for institutions to consider the role of the built environment as contributing to or detracting from experiences affecting mental health and wellbeing.

To date, there is little evidence regarding space on campus as it relates to student experience and mental health and wellbeing. Existing studies examining the influence of the post-secondary campus built environment on students' mental health and wellbeing seem to emphasize specific campus spaces (e.g., residences or classrooms) rather than looking at the campus more generally. A systematic review of the literature conducted by Evans & Stecker (2004) found that, by the early 2000s, a study conducted by Baum in 1977 was the most extensive study regarding chronic environmental stress and the concept of learned helplessness. In that study Baum looked at the impact of dormitory conditions on college students and found that conditions such as overcrowding impacted students' abilities to manage and regulate social interactions (as seen in Evans & Stecker, 2004). There are limitations in the Baum study. Overcrowding is just one aspect that could contribute to student wellbeing. Additionally, the living conditions represent just one environment that could contribute to poor mental health outcomes or academic performance and the current emotional state of the students prior to the study seems unknown. The foci of other studies in Evans & Stecker (2004) are more on the

environmental stressors and not on the participants per se. It is also worth noting that as of 2004 study, the most comprehensive research conducted on student mental health and wellbeing and the campus built environment is from 1977. The sparsity of research at the intersection of post-secondary students, the campus built environment, and student mental health and wellbeing is further evidenced in the results of the rapid scoping review that I conducted to address the first research question.

The demand for research exploring the relationship between people and the built environment is not new and, in fact, is core to an ergonomic-informed approach to design. While environmental ergonomics is a burgeoning field that looks at how humans respond to “social, psychological, cultural and organizational environments of systems” (Parsons, 2015, p. 642) it is typically viewed as emphasizing only the physical environment (Parsons, 2015). Broader inquiries into design and mental health may help inform our understanding of interactions and contexts (Eilouti, 2021; Trajković et al., 2021). One challenge facing the field of design is the lack of studies and established frameworks to guide research on this topic (Attaianese & Duca, 2012; Eilouti, 2021). Eilouti (2001) states, of buildings specifically, that there is a lack of recognition of them as facilitators of human-building interaction “and as an interface between users and the environment” (p.1). In my research, I aim to start to probe the interplay that occurs within an individual and with the built environment around them.

2.2 Stressors and coping in post-secondary students

Wellbeing is reflected in effective coping strategies. Coping requires people to assess a stressor (i.e., does this event/circumstance require action?) and reassess the situation to determine the appropriate response (i.e., how can the stressor be mitigated?) (Lazarus &

Folkman, 1987). Previous research findings show that environmental factors, such as housing quality, crowding, noise, and air quality are stressors that influence mental health (Evans, 2003; Evans & McCoy, 1998). When combined with psychosocial stressors (e.g., grief, family dynamics, peer pressure), mental health can become more precarious. Given the stressors that students are often experiencing, particularly as they transition to post-secondary education, gaining further insights into what supports or creates barriers in their coping strategies warrants investigation.

Stressors can also take the form of societal constructs (e.g., institutional policies, infrastructure, government programs) that create greater chasms of inequality, prolonging "socially patterned exposure" (Krieger, 2012, p. 936). Continued exposure to pervasive, uncontrollable, and unpredictable stressors may result in poor psychological outcomes such as increased agitation, anxiety, depression, learned helplessness, and emotional exhaustion (Codinhoto et al., 2009; Evans, 2003; Sullivan & Chang, 2011), thereby undermining physical, cognitive, social, and mental wellbeing (i.e., health) (Anisman, 2009; Raphael et al., 2020). This is to say that the stressors being experienced by students may go beyond circumstances in their personal lives and are in fact influenced by how the social systems, including institutions, are organized, the programs or services offered, and the infrastructure provided.

The transactional model of coping (Folkman et al., 1986) presents coping strategies as one mechanism that help individuals to manage or adapt to stressors. There are many ways of conceptualizing coping strategies (Freire et al., 2020). This project proposes five coping strategies or processes (Figure 2) (Aldwin & Yancura, 2004; Anisman, 2016). There is general acceptance that coping strategies can be classified as *problem-based* (e.g., reframing, breaking down into more manageable components), *emotion-based* (e.g., venting or containment of

emotions, ruminating, blaming self or others), or *avoidance-based* (e.g., distracting or disengaging from the stressor or stressor situation) (Anisman, 2014, 2016; Saleh Baqutayan, 2015). In addition to these three strategies, I include *social support seeking* and *religion-based strategies*.

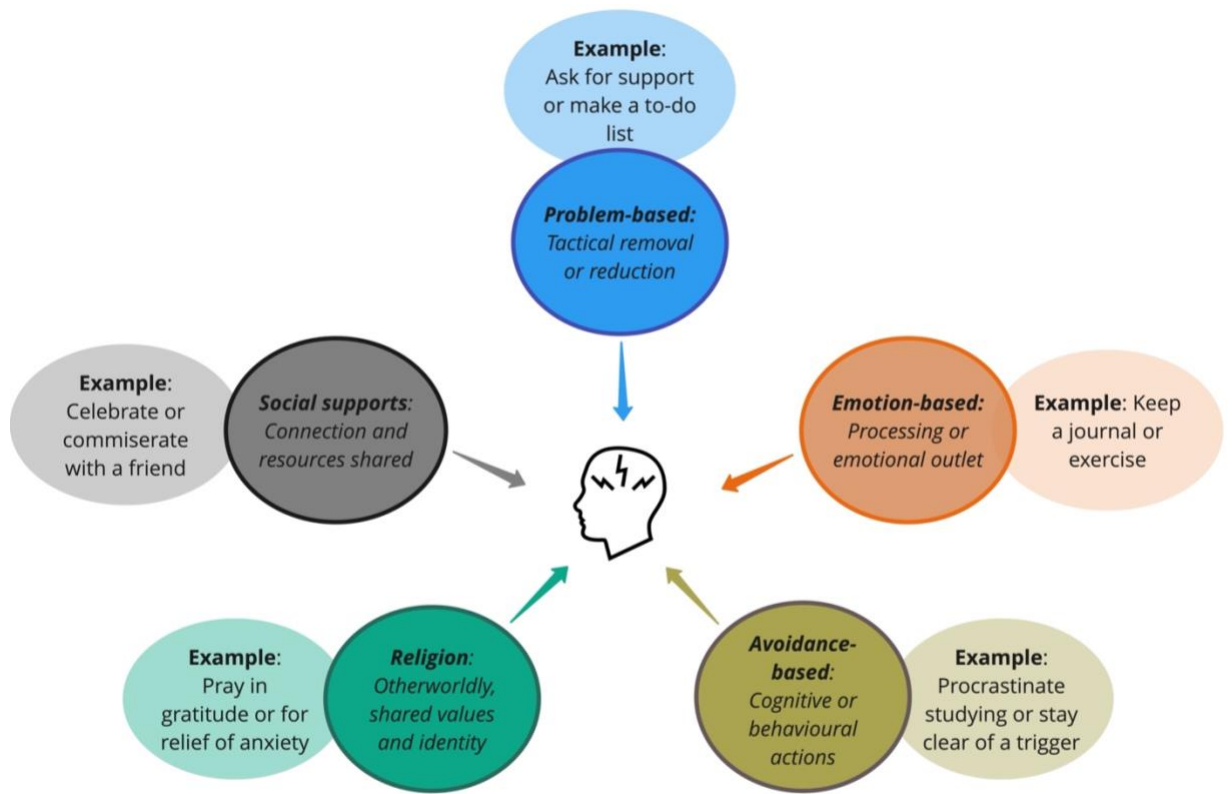


Figure 2 - Five categories of coping strategies (Aldwin & Yancura, 2004; Anisman, 2016)

Social support can play a vital role in a student's sense of belonging and experiencing feelings of appreciation (Alsubaie et al., 2019). Transitioning to post-secondary education is a vulnerable time for many young adults as they deal with interconnecting demands associated with academic stress, social pressure, and interpersonal challenges. For instance, some students move away from home and live on campus in residence. Depending on the approach of the institution or the academic program, these same students could spend the next three to four years

studying and living together, with little time or space to distance themselves. There is less reliance on family support as the students increase their independence and autonomy while exploring social identities (Park et al., 2020). Research demonstrates that social supports from professors, peers, and family are important during this time and key to aiding in the transition to post-secondary education (Urquhart & Pooley, 2007).

Spirituality and religion may have an effect on a person's mental health and wellbeing. *Religion-based* strategies are a complex coping resource drawing on elements of motivation and purpose, avoidant behaviours, and seeking of social connection and belongingness (Ysseldyk et al., 2010). When people subscribe to or practice the idea of a higher order, they tend to employ other adaptive coping strategies at the same time (Fischer et al., 2010; Ysseldyk et al., 2010). In the early to mid 2000s, research documented a revival of religiosity on post-secondary campuses and its positive affect on levels of satisfaction and academic performance (Waithaka & Gough, 2017). Students who exuded religious commitment tended to study more, engage in extracurricular activities, and party less than other students (Waithaka & Gough, 2017). Given these findings, I include religion-based strategies as another coping resource to learn more about students' experience with the campus built environment.

While coping strategies are often classified as healthy versus unhealthy it is not an either-or situation (Anisman, 2016; Haslam et al., 2018). Similarly, individuals may draw on a combination of strategies that are dependent on predisposing individual differences, available resources (e.g., social support), the context, as well as features of the stressor itself (Matheson et al., 2020). Thus, coping strategies represent just one approach to look at how students respond to the stressors in their lives in both healthy and unhealthy ways.

2.3 Emotions as a motivation to understanding experience of the campus built environment

There are many approaches to studying coping resources. This research explores students' personal experiences based on emotions and campus spaces regardless of specific stressors perceived to be positive or negative that they may experience as part of student life. By focusing on the emotional experience of students within the campus built environment, the student's personal experience is at the centre of the research.

Studying emotions in relation to mental health and wellbeing is not novel. There are links between stressors and emotional experiences (Shuman & Scherer, 2015) and how emotions help or hinder navigation of daily life events (Zautra et al., 2005). Emotions are indicators of how we react, respond, and cope with stressors (Zautra et al., 2005). Early research in basic emotion theory posits that there are a finite number of basic emotions that makeup the human experience (e.g., happy, sad, afraid, anger, surprise, and disgust) (Ekman, 1992). In recent years, there has been much debate as to whether our emotional experiences are limited to this set of six emotions. For instance, one study indicates there are more than two dozen different types of emotions (Cowen & Keltner, 2017). Yet, it can be argued that perhaps there is a core set of emotions that branch off into other emotions, the interpretation of which can be dependent on personal experiences. For instance, Ekman (1992) presents happiness or joy as one of the basic emotions and while this, too, is reflected in Cowen & Keltner (2017), amusement is indicated as a separate emotion. The determination of whether the emotion experienced is in fact amusement or happiness is driven by personal experiences and interpretation. I opt to focus on the six basic emotions presented by Elkman as they provide a broad foundation on which to understand better

human emotions across the population of students participating in the research. Additionally, given the scope of my research, the limited list will facilitate the identification of commonalities and differences in emotional experiences across individuals.

2.4 Salutogenic theory: A guiding theory

Salutogenic theory (developed by social learning theorist Aaron Antonovsky (1979)) offers a model that links a person's generalized coping response to situational events by focusing on what builds the continuum of ease/dis-ease rather than health/illness (Patuano et al., 2022). In shifting the perspective in this manner, salutogenesis presents an asset-based approach to understanding stress. A key asset in the salutogenic model of health is a "sense of coherence".

Our sense of coherence seems to develop during early adulthood (Brett et al., 2022), making it important for this research given that this is when a lot of youth transition to post-secondary education. A sense of coherence is a global orientation of self (i.e., a way of seeing the world) that is the ultimate resource through which a person appraises and responds to stressors (Antonovsky, 1979; Vinje et al., 2017). Thus, a sense of coherence is a resource that helps people cope and breaks down into three main domains: *comprehensibility* (e.g., how we make sense of life and the world around us), *manageability* (e.g., how we cope and our behavioural responses to circumstances), and *meaningfulness* (e.g., how we get motivated or purpose in life) (Brett et al., 2022; Golembiewski, 2010; Hochwlder, 2019). While a sense of coherence uses cognitive and behavioural responses to circumstances and events, it does not seem to explain or account for how people gather and create emotional meaning. The lack of account for emotional meaning as part of the sense of coherence motivated Karol & Smith (2021) to present a fourth domain to support salutogenesis, namely that of *emotional affordance*. Emotional affordance, as

presented by the researchers, represents the ability of an environment to promote a cohesive emotional reaction by meeting an individual's social and psychological requirements and offering opportunities for the development of a sense of identity, pleasure, and agency (Karol & Smith, 2021).

In the salutogenic model of health, a person experiencing ease has a healthy sense of coherence where they interpret and react to stressors using more adaptive coping strategies. We may need to give consideration, though, to how the sociocultural context, physical environment, and human health influence stress and disease processes (Vinje et al., 2017). For instance, we may need to consider that life circumstances that go beyond an individual's control may affect their sense of coherence. Salutogenic researchers such as Vinje, et al. (2017) and environmental psychologists and architects such as Evans (2003), Shepley, et al. (2017) and Vischer (2008a) make consideration for this in their research. It is important to recognize these elements because the ease of health experienced by individuals is not fully within their control and is dependent on other factors such as race, gender, disability, access to healthcare, and living wage (i.e., social determinants of health).

The salutogenic model of health brings forward two key concepts that inform our sense of coherence and that pertinent to this research: generalized resistance resources (GRRs) and specific resistance resources (SRRs) (Figure 3). General resistance resources represent "characteristics of a person, a group, or a community that facilitate the individual's abilities to cope effectively with stressors" (Mittlemark, 2017, p.57) including biological, cognitive, interpersonal, and attitudinal factors. Specific resistance resources can be "optimised by societal action" (Mittlemark, 2017, p.71) and are external to any one individual, reflected in social and physical environments such as programs, services, and infrastructure. Having a 'sense of

coherence’ helps us mobilize these resources, ultimately aiding in our ability to manage stress by providing a framework for understanding and making sense of challenging situations (Eriksson & Lindström, 2007). It enables individuals to perceive stressors as less threatening and more manageable, fostering a more adaptive response. People with a higher sense of coherence may be more likely to employ effective coping strategies, such as problem-solving, seeking social support, and engaging in self-care activities (Eriksson & Lindström, 2007). Collectively, GRRs and SRRs contribute to a person’s sense of coherence which guides them on the ease/dis-ease continuum (Mittlemark et al., 2022).

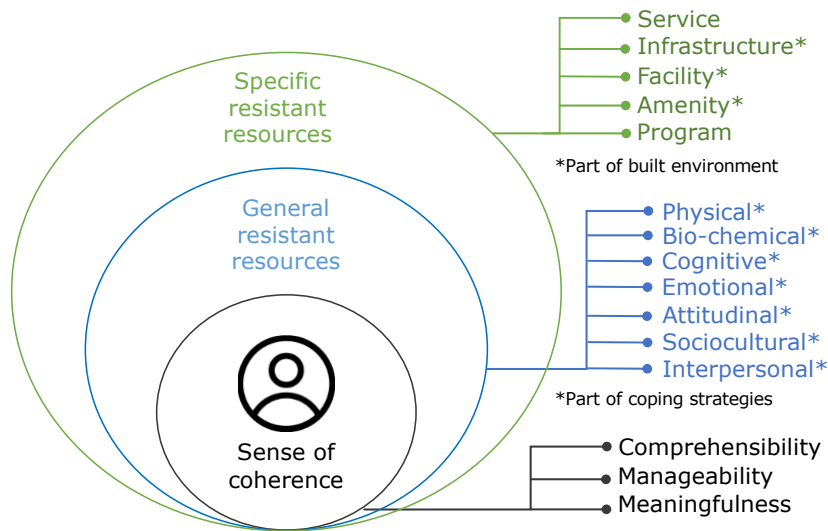


Figure 3 - Relationship of how the built environment interacts with coping strategies and aspects of salutogenesis (Erikson, 2016; Mittlemark, 2017).

Over the course of his career, Antonovsky was critical of the attention that researchers paid to the sense of coherence as a cause of ease/dis-ease (i.e., independent variable), rather than the effect of ease/dis-ease (i.e., dependent variable) (Mittlemark et al., 2017). Additionally, a significant portion of the literature about salutogenesis and health focuses on GRRs in isolation

from SRRs (Vinje et al., 2017). Yet, SRRs could be mediators of the sense of coherence but a lack of research in this area persists (Mittlemark et al., 2017). The complex and compounding factors that make up SRRs such as physical space, services, and policies (i.e., SRRs) and, how or if, they affect GRRs may be difficult to study and replicate.

Both GRRs and SRRs may diminish or promote tensions we interpret as stress (Idan et al., 2017; Mittlemark et al., 2017). The research proposed here explores aspects of the campus built environment as an SRR and its potential influence on emotions and coping strategies (i.e., GRR). By blending core concepts from earlier research on stress, coping, and emotion with those from salutogenesis, this research project centres its approach on the students' emotions and experiences in relation to their experience of campus built environments and how this contributes to their mental health and wellbeing.

2.5 Summary

The primary focus of this research is the exploration of students' experiences with the campus built environment in the context of their mental health and wellbeing. Guided by specific components of salutogenesis (a theoretical framework that focuses on the factors that promote health and wellbeing), I explore student experiences in campus spaces through the constructs of coping strategies and emotions. Collectively, coping strategies, emotions, and salutogenesis represent a multi-dimensional approach to understanding the interplay and nuances in how various aspects of the built environment may impact student mental health and wellbeing.

Chapter 3: Research Question 1 – Methods, Results, and Discussion

3.1 Methods

My first research question asked, “*within the literature, what relationship, if any, exists between the post-secondary campus built environment and students’ mental health and wellbeing?*” Results of a rapid scoping review demonstrated that the built environment on post-secondary campuses may be linked to students' mental health and wellbeing, though studies in the past ten years remain sparse in peer-reviewed academic literature. This study followed the five stages of rapid scoping review by Arksey and O'Malley (2005), outlined below.

1. Identify the research question

Post-secondary institutions represent an important environment in which to explore mental health. The primary research question was: “*Within the literature, what relationship, if any, exists between the post-secondary campus built environment and students' mental health and wellbeing?*”.

2. Identify the relevant studies

A list of search parameters captured the relevance of possible terms and experiences (Figure 4). The intersection of four subject/topic areas was of particular interest, namely built environment, mental health, postsecondary educational environments, and students. All four topics had to be present to include a particular article in the review. The multidisciplinary nature of the topic and potential disciplines involved lent to extensive searches across various databases, including PubMed, Scopus, CINAHL (Cumulative

Index to Nursing and Allied Health Literature), PsycInfo, Art and Architecture, ERIC (Education Resources Information Center), and ASSIA (Applied Social Sciences Index & Abstracts). I created a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) diagram, presented later in this section, to share results.

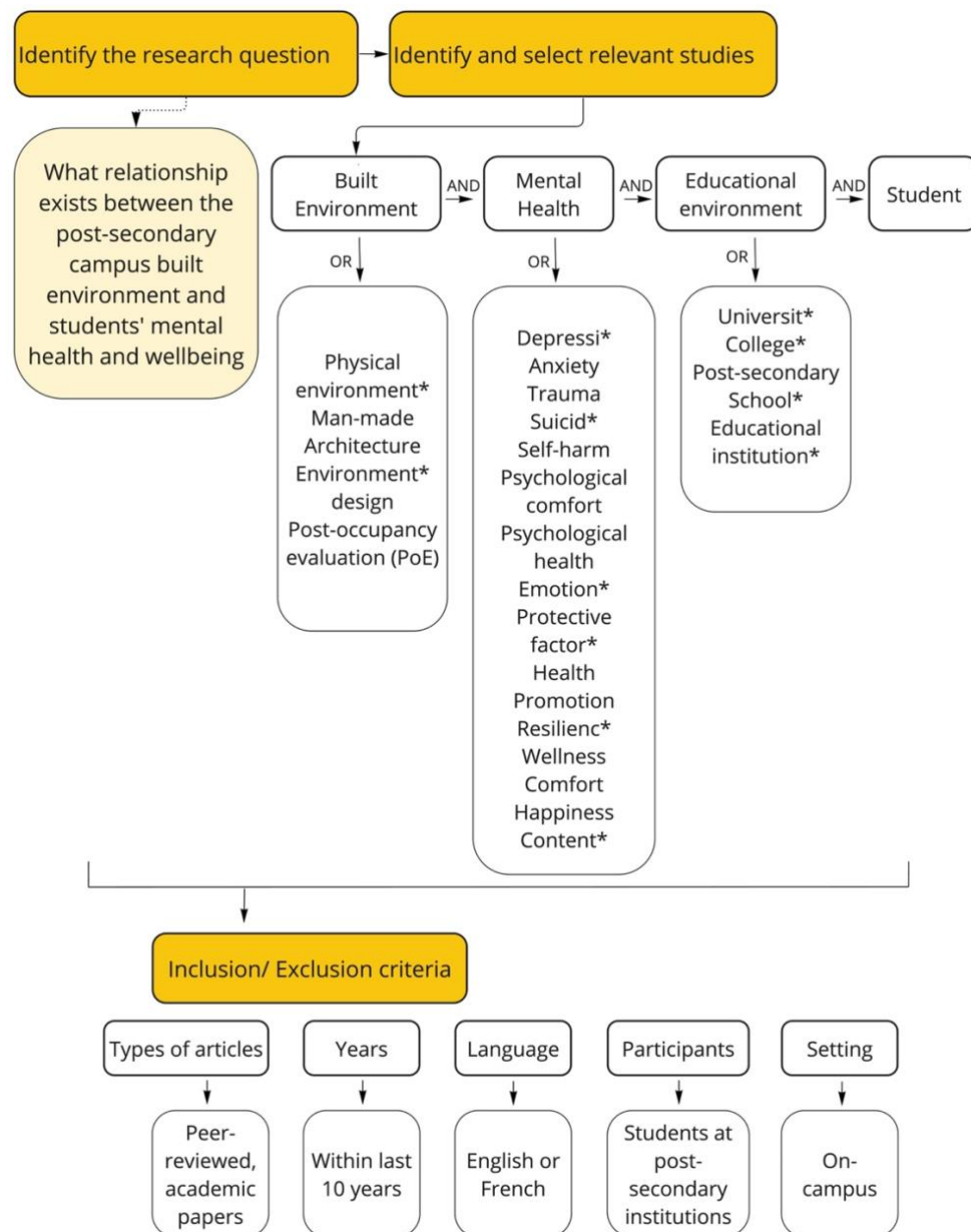


Figure 4 - Flow chart capturing the design of the rapid scoping review

3. Study selection

The narrow scope of the research question as well as limitations of resources, informed the study inclusion criteria:

- a. *Type of articles*: The types of articles included international, peer-reviewed academic papers representing the study of campus spaces and student perspectives.
- b. *Language*: Although English is my primary language, a working knowledge of French facilitated inclusion of studies in either language.
- c. *Date of publication*: I included studies since 2011. I justified this decision as there have been sociocultural shifts influencing the current mental health crisis; a crisis that involves increasing amounts of psychological distress, major depression, or suicidal thoughts (Twenge et al., 2019).
- d. *Population and location of the built environment*: The population of interest was post-secondary students from any country.

After compiling all the research for study eligibility and deciding which studies would remain, a two-phase screening process began. The first phase included a review of titles and abstracts, while full text reviews took place in the second phase.

4. Charting the data

As part of the review, I captured and charted relevant data included in Table 2. This included selected publications focused on aspects of the inclusion/exclusion criteria. Additional information included study-specific data such as *research objectives* identified in the article, *population studied*, the *type and location of the built environment* examined

as part of the research, pertinent *theories* used to anchor the research, *study design/methods*, *analysis methods*, and the *summary of findings*, including recommendations.

Table 2 - Rapid scoping review articles included in study

Author	Title	Research objective	Theories mentioned	Study design	Analysis methods	Summary of findings
Bowrey, David J., Kidd, Jane M.	Explore emotional experiences in the operating theatre of medical students	Understand emotions experienced by students when in the operating room for the first time Impact of emotions on students' learning	none	Email recruitment Semi-structured interviews, transcribed and coded	Qualitative themes of emotions	All participants reported initial negative emotions driven by loss of familiarity, organizational issues, concerns about violating protocol, and a fear of fainting Learning facilitated after positive emotions once familiar with activity
Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.	Influence of the residential environment on undergrad health	Estimate prevalence of different diseases by demographics Estimate the relationship between a dorm's characteristics and related disease incidence	none	De-identified billing codes from private university clinic linked to students' residential history and demographic information	Two-stage, hierarchical regression model	Significant differences between student levels when it came to the prevalence of mental health, upper respiratory infections, and substance abuse Differences noted depending on type of ventilation and room size

Author	Title	Research objective	Theories mentioned	Study design	Analysis methods	Summary of findings
Licite L., Janmere L.	Student expectations towards physical environment in higher education	Aim is to examine students' expectations towards the physical environment in higher education	none	Survey expectations towards the physical environment using semantic, differential and expectation indicators	Central trend and dispersion indicators Non-parametric tests (Mann-Whitney test)	Opinions showed homogeneity The ratings of some aspects represented statistically significant differences
Marchand, Gwen C., Nardi, Nicholas M., Reynolds, Douglas, Pamoukov, Stoil	The impact of the classroom built environment on student perceptions and learning	Study the influence of light, sound, and temperature on class performance	Bioecological model of human development	Study of classroom tasks in lab environment using sensors, meters Survey of experience	Positive Negative Affect Scale (PANAS) Exploratory Factor Analysis (EFA) using principal axis factoring (PAF) to identify possible subscales 2x2 factorial ANOVA	Participants in the listening condition had lower scores on a comprehension test
Peters T., D'Penna K.	Biophilic design for restorative university learning environments: A critical review of literature and design recommendations	Examine research from numerous fields to understand biophilic design benefits for mental health, creativity, improved output, and more comfortable spaces for learning	Place Attachment, Restorative Design Environmental Design Attention Restoration Stress Reduction Prospect-Refuge	Semi-structured literature review	Analysis and categorizing of studies	Evidence-based design recommendations for biophilic design in university settings

Author	Title	Research objective	Theories mentioned	Study design	Analysis methods	Summary of findings
Reese R.F., Seitz C.M., Gosling M., Craig H.	Using photovoice to foster a student vision for natural spaces on a college campus in the Pacific Northwest United States	Examine if campus greenspaces play a role in preventing stress and promoting student health	none	Community-based, participatory research	Thematic analysis of photos	Five student-created themes emerged: mental and physical health; developing community spaces; sustainable infrastructure design; preserving habit and history and; issues related to homelessness
Yildirim K., Capanoglu A., Cagatay K.	The effects of physical environmental factors on students' perceptions in computer classrooms	Subtle differences in interior design/layout (rather than larger matters of architectural design and floorplans) assumed to influence the perceptual performance of design students	none	Survey using Likert-style scale for students to respond to modifications made in learning space	Alpha coefficients of dependent variables to determine whether there were relationships between the dependent variables One-way ANOVA to examine the effects of the differences in interior design elements	Environmental settings had an important influence on the perceptual evaluations Participants perceived space positively or negatively depending on proximity to equipment or people, presence of aesthetic features, and circulation sightlines

Table 2 - Rapid scoping review articles included in study

5. Collating, summarizing, and reporting

I conducted deductive thematic analysis by reviewing the collected data and then coded it based on direct influences (e.g., housing, crowding, noise, indoor air quality, and light) and indirect influences (e.g., personal control, social support, and restoration) of the built

environment (Evans, 2003). I then applied the same technique based on the theoretical framework of salutogenesis, more specifically, salutogenic resources (e.g., general resistance resources and specific resistance resources (Mittelmark, 2017). The analysis helped me to find common elements and outliers among the articles (Saldaña, 2016). I colour-coded each author's name and any key insights from the article and mapped those to column headings that represented characteristics I looked for in each study (see Appendix A).

3.2 Results

As indicated in the PRISMA chart (Figure 5), the initial search on the topics yielded 129 papers. As part of the preliminary screening, I removed duplicate records (n=26), articles that were not in English or French (n=1), and articles that were not relevant to the topic at all (n=7). I then screened the title and abstracts of the remaining articles (n=95) to see if they represented the intersection of the topics. Of those reviewed, I excluded articles not relevant to post-secondary institutions (n=14), built environments at post-secondary institutions (n=44) (e.g., referred to educational institutions more broadly), or built environment at post-secondary institutions and student mental health (n=13). This left a total of 24 articles on which I would conduct a full screening. After the full screening process, I determined that an additional 16 of the remaining 24 articles were excluded due to incorrect context (e.g., built environment referring to an academic program rather than infrastructure or surroundings). This process left me with eight of the original 129 articles for full review and analysis.

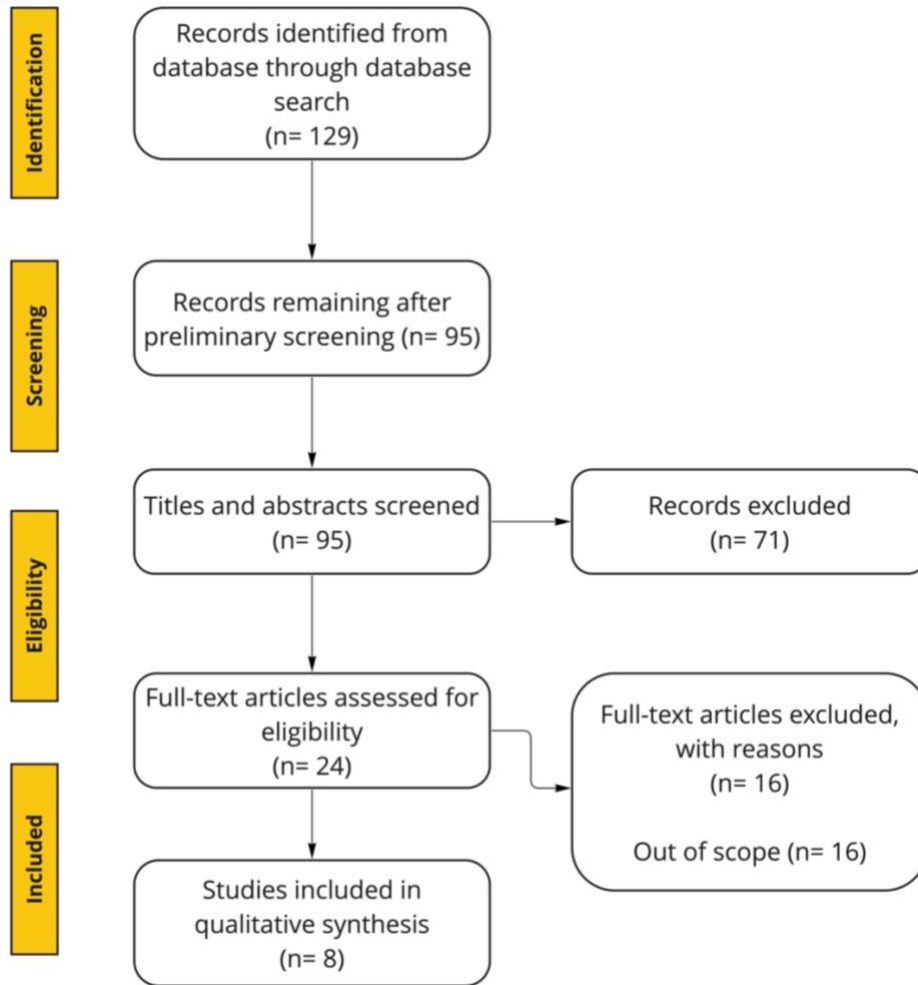


Figure 5 - PRISMA diagram documenting the primary stages of the rapid scoping review

The research in the eight articles in the review took place in five different countries (Turkey (n=1), Latvia (n=1), Canada (n=1), England (n=2), and the United States (n=3)) in the past decade. The studies represented six disciplines, including: *architecture* (Peters & D’Penna, 2020; Yildirim et al., 2020), *education psychology* and *counselling* (Marchand et al., 2014; Reese et al., 2020), *teaching and education* in medicine (Bowrey & Kidd, 2014), *environmental and public health* (Cedeno Laurent et al., 2020), *environmental design* (Foo & Mavrogianni, 2018), and *sociology* (Licite & Janmere, 2018). This was reflective of the breadth of disciplines and

approaches researching the built environment in the literature. While there was a variety of methods and analytic techniques used (e.g., semi-structured interviews with thematic coding, sampling with ANOVA tests), none of the studies included interventions, although suggestions for future directions called for interventions (Table 2).

In addition to charting categories reflected in Table 2, I added three additional categories to chart based on the literature in Chapter 2: Background. These categories included: 1) the *types of campus environments studied*; 2) *direct and indirect influences of the built environment* (as defined by Evans, 2003) *on post-secondary campuses*; and 3) *studying the sense of coherence by looking at general and specific resistance resources* (as defined by Erikson, 2016; Mittlemark, 2017). I collated the results shown in Table 3.

<i>Table 3 – Additional characteristics of studies in scope</i>				
Authors	Title	Type of environment	Influence of the built environment*	Salutogenic resources**
Bowrey, David J., Kidd, Jane M.	How do early emotional experiences in the operating theatre influence medical student learning in this environment?	Learning environment	Indirect influences (emotional experiences)	Specific resistance resource (operating theatre infrastructure)
Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.	Influence of the residential environment on undergraduate students' health	Living environment	Direct influences (air and thermal quality, housing, crowding)	Specific resistance resource (university residences)
Foo J., Mavrogianni A.	Seeing is believing, or is it? An assessment of the influence of interior finish characteristics on thermal comfort perception at a University campus in a temperate climate	Learning environment	Direct influences (thermal comfort perception)	Specific resistance resource (university lecture theatres)
Licite L., Janmere L.	Student expectations towards physical environment in higher education	Learning environment	Direct influences (layout and size of study rooms, ergonomics, and technologies)	Specific resistance resource (spaces where students study)

Authors	Title	Type of environment	Influence of the built environment*	Salutogenic resources**
Marchand, Gwen C., Nardi, Nicholas M., Reynolds, Douglas, Pamoukov, Stoil	The impact of the classroom built environment on student perceptions and learning	Learning environment	Direct influences (light, sound, and temperature) Indirect influences (mood and cognitive function)	General resistance resource (student mood and cognition) Specific resistance resource (classroom)
Peters T., D'Penna K.	Biophilic design for restorative university learning environments: A critical review of literature and design recommendations	Learning environment	Indirect influences (emotional stress, happiness, stimulation, cognitive function, social support, belonging)	General resistance resource (Emotional stress, happiness, cognitive function of students) Specific resistance resource (Environmental and biophilic design provided by university)
Reese R.F., Seitz C.M., Gosling M., Craig H.	Using photovoice to foster a student vision for natural spaces on a college campus in the Pacific Northwest United States	Outdoor space	Indirect influences (restoration from stress or fatigue)	General resistance resource (student health)
Yildirim K., Capanoglu A., Cagatay K.	The effects of physical environmental factors on students' perceptions in computer classrooms	Learning environment	Direct influences (layout, acoustics, climate control, colour and lighting)	Specific resistance resource (computer classrooms at university)

Table 3 - Additional characteristics of studies in scope

*Defined by Evans (2003)

**Defined by Erikson (2016) and Mittlemark (2017)

3.2.1 Types of campus environments

In reviewing the articles, several areas of the campus built environment emerged as the focus of study designs. The majority of the studies (n=6) focused on learning environments.

There was uniformity in how learning environments were interpreted by the authors, comprising

formal learning settings such as a classroom, theatre, or laboratory. One study explored living environments, namely university residences/dormitories. Another study focused exclusively on the outdoor environment, in this case, defined as natural spaces that were within the control of the university. Each study constrained itself to a specific campus environment with none of the studies looking at campus environments as a whole. While I did not notice any significant variations in the findings, I noticed that regardless of the environment being studied, there were similarities. For instance, Yildirim et al. (2011) and Bowrey et al. (2014) results indicated that the environmental settings had a potential influence on how students evaluated the environment, whether the learning environment was an operating theatre or a science laboratory. Reese et al. (2020) and Peters et al. (2020) both discussed the importance of nature with the former pointing to the need for biophilic design indoors and the latter calling for institutions to create community-based greenspaces as part of student experiences.

3.2.2 Direct and Indirect Influences of the Built Environment

In Chapter 2, I noted that Evans (2003) highlighted both the direct and indirect influences that the built environment could have on mental health. Direct influences included environmental characteristics such as crowding, noise, indoor air quality, and light (Evans, 2003a). Yet, characteristics of the built environment can lead to indirect influences on an individual including personal control, socially supportive relationships, and restoration from stress and fatigue (Evans, 2003b).

Accordingly, I charted three studies that focused primarily on *indirect influences* with key aspects involving emotional processing, prevalence of mental health issues, or stress prevention (Bowrey & Kidd, 2014; Peters & D’Penna, 2020; Reese et al., 2020). Four studies focused on *direct influences* of thermal comfort, light/sound/temperature, and general design and

layout on student mental health (Cedeno Laurent et al., 2020; Foo & Mavrogianni, 2018; Licite & Janmere, 2018; Yildirim et al., 2011). One study looked at both indirect and direct influences of the built environment on student mental health in terms of whether combined environmental factors (e.g., light, sound, temperature) could impact student mood and learning (Marchand et al., 2014).

Two of the studies looking at direct influences on the built environment (Cedeno et al., 2020; Licite & Janmere, 2018) introduced the idea that there could be different responses depending on a student's level of study (e.g., freshman versus upperclass) or discipline of study (e.g., social science versus science and engineering). For student level, the results may have been influenced by the type of dormitories assigned to the students as well as the socialization differences between freshmen and later years (Cedeno et al., 2020). Licite & Janmere (2018) noted that while the opinions shared as part of the research demonstrated homogeneity, statistically significant differences did show up among disciplines which could be attributed to the types of learning spaces used in different academic disciplines.

3.2.3 General and specific resistance resources

General resistance resources (GRR) and specific resistance resources (SRR) showed up in the initial literature review as aspects of a potential unifying theory (i.e., salutogenesis) that could have an influence on students' coping strategies, and subsequently their mental health and wellbeing. As a reminder here, GRRs include physical, bio-chemical, cognitive, emotional, attitudinal, sociocultural, or interpersonal resources, whereas SRRs refer to services, infrastructure, facilities, amenities, or programs (Mittlemark et al., 2017).

Table 2 conveyed the different campus environments in relation to whether the focus of the study was primarily GRRs, SRRs, or both. Two studies looked at both GRRs and SRRs,

namely Marchand et al. (2014) and Peters et al. (2020). Marchand et al. (2014) examined the impact of the classroom built environment on student learning. In Marchand et al. (2014), GRRs that may have influenced student coping strategies included student mood, reaction to stress, and cognition. The classroom represented the SRR. Peters et al. (2020) identified emotional stress, happiness, and cognitive function as GRRs that influenced student coping. The SRRs included the environmental and biophilic design. Out of the eight studies, one study focused solely on GRRs. Reese et al. (2020) explored stress prevention and student health. Participants used photovoice as a method to craft a vision for how natural spaces on campus could be minimally developed as a means to promote student health. In the case of this study, student health represented a GRR. It was unclear based on the article if the natural spaces talked about were in control of the institution and purposely built, therefore I did not code the space as an SRR.

The remaining studies primarily focused on SRRs. Bowrey et al. (2014) focused on the adjustment of medical students experiencing an operating theatre environment for the first time, including the physical design of the space and students' feelings of social inclusion. Cedano et al. (2019) as university residences, Foo et al. (2018) as lecture theatres, Licite et al. (2018) as study spaces, and Yildirim et al. (2011) as design and layouts of a computer classroom.

I noticed that research that highlighted SRRs specifically noted a potential relationship or influence between the built environment and learning outcomes. For example, Marchand et al. (2014) noted the role that a substandard learning environment can play in negatively impacting an adult's capability to learn and perform well. Yildirim et al. (2011) found that the interior design elements in learning environments may influence student motivation and concentration. I saw similar conclusions reflected in Licite & Janmere (2018) who argued that the physical

environment at an institution can impact the study process of students, with ideal characteristics including advanced technology and personal comfort.

The rapid scoping review supported the initial impression I had, which was that research at the intersection of the built environment on post-secondary campus and student mental health was sparse. The studies that did exist, did not look at how or why students use the campus environment as guided by their experiences but, instead, pre-determined the type of environment studied. Furthermore, the studies primarily examined GRRs and SSRs in isolation of each other.

3.3 Summary

In the first study, I conducted a rapid scoping review to explore the intersection between the built environment of post-secondary campuses and student mental health. The studies I reviewed primarily focused on learning environments like classrooms, theaters, and laboratories. The articles also discussed the direct and indirect influences of the built environment on mental health, including environmental characteristics such as noise, temperature, and lighting.

The rapid scoping review revealed that research at the intersection of the built environment on post-secondary campuses and student mental health was lacking. Additionally, the studies mostly examined general resistance resources (e.g., student health, emotional coping) and specific resistance resources (e.g., classroom, laboratory) in specific campus environments, without considering campus environments as a whole. Based on the findings, the need for more research to explore how students experience the campus environment more broadly seemed an appropriate next step.

Chapter 4: Research Question 2 – Methods, Results, and Discussion

4.1 Methods

The goal of the second study was explore the following question: *Are coping strategies and design features on campus related to students' considerations of where to go when experiencing different emotions?* The primary method used in this study was a questionnaire (Appendix B). I structured the questionnaire into four primary sections, including three research sub-questions as well as questions to collect demographic information (Figure 6).

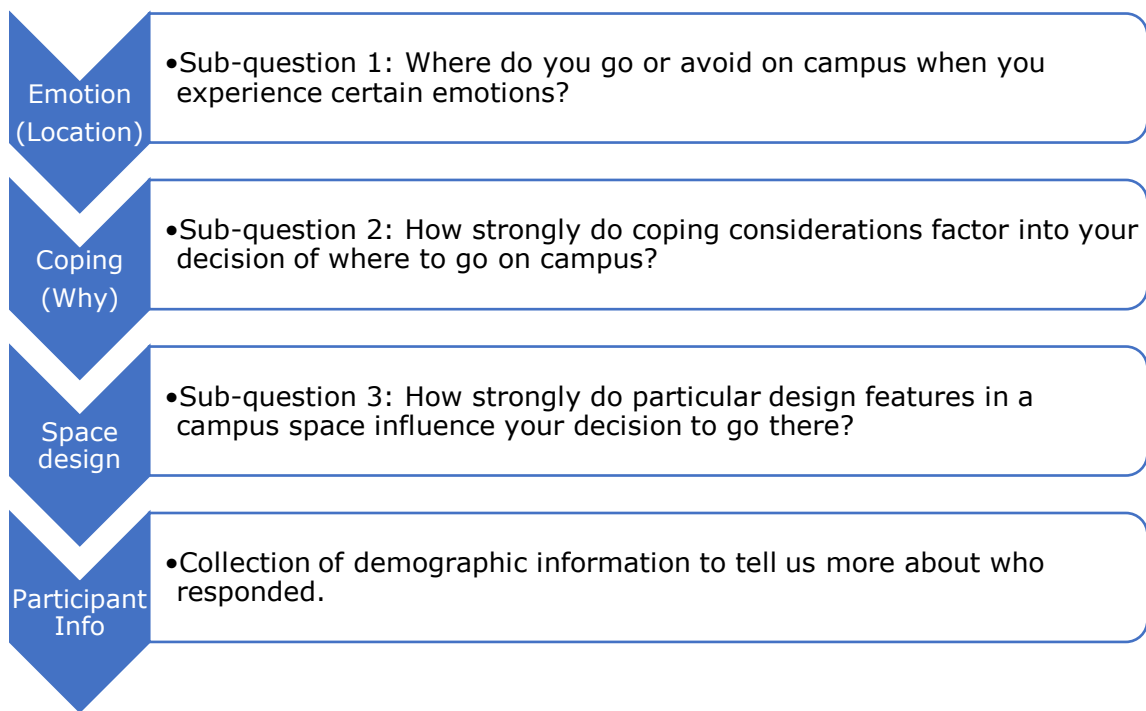


Figure 6 - Process and design of the questionnaire for the research study

1. *Sub-question 1: Where do you go or avoid on campus when you experience certain emotions?*

I introduced six separate emotion statements to participants (i.e., you are happy, sad, afraid, angry, surprised, or disgusted) to which they responded where they would go or where they would avoid on campus when they felt this way. For example, “*You are feeling happy or joyous. Where do you go on campus? Please name one specific space on campus in the box below.*” I asked the respondents to be as specific as possible when naming a space and included a campus map (Appendix C) for each question to assist them in responding. Students then entered their answer in a text box. This process repeated for each of the six emotions.

2. *Sub-question 2: How strongly do coping considerations factor into your decision of where to go on campus?*

I explored why students may use certain spaces on campus as part of coping. Specifically, I asked them about their level of consideration when deciding to go to a space based on a list of pre-determined coping strategies. These strategies included hanging out with friends, releasing emotions, avoiding emotional triggers, accessing resources for support, addressing spiritual needs, or other considerations. By asking this question, I hoped to gain initial insights into why students use certain spaces on campus as part of their coping mechanisms.

Students rated the consideration they would give to each reason on a scale that ranged from little (coded 1) to strong (coded 3) consideration. The reasons to consider spaces mapped to coping strategies (although students did not know this when completing the questionnaire) which included *social support, emotion-based, avoidance-*

based, problem-based, and religion-based strategies (Figure 7). Students could select ‘not applicable’ should they feel that the coping strategy did not apply in the emotion statement.

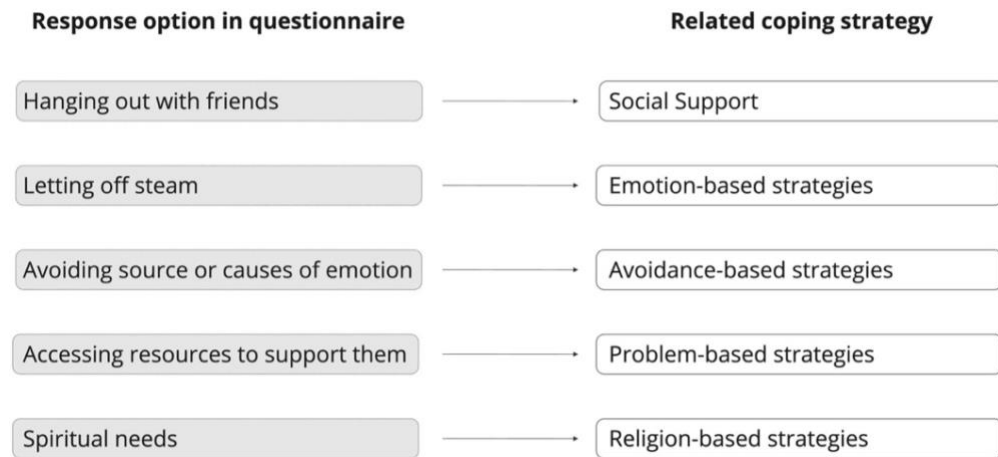


Figure 7 - Mapping of response options in questionnaire to related coping strategies

3. *Sub-question 3: How strongly do specific design features in a campus space influence your decision to go there?*

To gain insight into how students might react to different design features, I inquired about their perceptions of how much certain design features influenced their decision to be in a particular space. This was in reference to their previous responses about where they would go when experiencing different emotions. The third sub-question specifically asked about the self-reported influence of specific design features of the campus space they had previously reported going to when experiencing each emotion.

Respondents adjusted a slider for thirteen distinctive design features to indicate how strongly a particular feature in the space may influence their decision to go there. The slider ranged from 0 (feature had no influence) to 100 (feature had the strongest influence), increasing in increments of ten. Respondents could also share other features

not included in the original list. There was a *not applicable* option should a particular feature not apply.

4. *Collection of demographic information:*

I collected demographic information to gain a better understanding of participants that responded to the questionnaire. I used checklists with pre-determined options for *age*, *current level of study*, *year of study*, *first-generation enrollment*, and *disability*. I left *ethnocultural identity* and *gender* as open-ended text boxes. I asked students to please list all relevant groups using their own words.

The study received clearance from the Carleton University Research Ethics Board (clearance #117115).

4.1.1 Recruitment

Recruitment for student participants took place in March 2022 and occurred primarily through emails sent to departmental administrators across campus, outreach to student services and societies, and through personal campus contacts. I also recruited students in the business school through Sona, a research participation tool. I shared instructions that outlined the purpose of the study as an exploration of what spaces students would go to or avoid on campus when they were feeling particular emotions. The questionnaire closed on April 8, 2022. I informed participating students that the online questionnaire would take 20-30 minutes. To be eligible to take part, students needed to be:

- Eighteen years of age or older.
- Able to read and write in English.
- A current undergraduate or graduate student at Carleton University with a valid Carleton email address; and

- Familiar with and have experience of Carleton's physical campus.

Carleton student participants who completed the questionnaire could enter a lottery for a chance to win a \$50 gift card. I selected one name for every twenty-five participants in the lottery until the questionnaire closed on April 8, 2022.

4.1.2 Analysis

For the first sub-question (“*Where do you go or avoid on campus when you experience certain emotions?*”), the first step in the data analysis process involved cleaning the dataset and creating consistent naming conventions (Saldaña, 2016) for locations identified by participants. This was necessary as different students might refer to a specific location in different ways. For instance, one student may have said “Starbucks in the University Centre¹”, and another “UC Starbucks”. Following this step, I created two views of the cleaned data. The first view was the direct responses provided by the participants (e.g., University Centre Starbucks). Simply relying on this view of the data could result in me missing some patterns of use of campus spaces and infrastructure. For instance, relying solely on the students’ responses, a specific space named as “the Library” would be counted separately from a student who responded, “a quiet room on the fifth floor of the Library”. Thus, I created a second view of the data by rolling up each location to a campus building such as the MacOdrum Library (also known as the Library) or designated areas, like the outdoor space, the Academic Quad (also known as the Quad) or the series of buildings and spaces known as Athletics. Collectively, I refer to the categories of buildings and

¹ Two buildings referenced frequently in the study changed names during the course of the research (University Centre became Nideyinàn; Residence Commons became Teraanga Commons). Given that the older names were in place at the time of the research and that was the context in which participants answered questions, I have reflected the original names in the results and discussion.

designated areas as *campus spaces*. I then created word clouds of the campus spaces that student would go to or avoid when experiencing each of the six emotions.

Some students provided insights as to why they would go to or avoid a particular space when naming a location. For example, a student might state they would avoid the University Centre because of the busyness and crowds. I used MAXQDA 22, a qualitative analysis tool, to conduct deductive thematic analysis, using key words that could describe perceived direct and indirect influences of the built environment (Evans, 2003) and that repeatedly appeared in responses (Table 4).

<i>Table 4 - List of key words based on Evans (2003) that repeated in qualitative responses</i>		
Direct influences	Indirect influences	Repetition in text
Crowd* or People or Busy* or Alone or Deserted or Priva*	Social or People or Commun* Friend*	Ugly or Gross
Noise or Sound or Quiet or Loud or Traffic	Restore or Fatigue or Tired or Stress* Destress*	
Light	Control	
View or Out*		
Air		

Table 4 - List of key words based on Evans (2003) that repeated in qualitative responses

For sub-question 2 (*How strongly do coping considerations factor into your decision of where to go on campus?*) and sub-question 3 (*How strongly do particular design features in a campus space influence your decision to go there?*), I used the Statistical Package for Social

Sciences (SPSS) software platform to conduct preliminary descriptive analyses (frequencies and means) which I, then, visualized using heat maps. For sub-question 2, I assigned a numerical value to each level of consideration for each coping response in SPSS (*strong consideration* = 3, *moderate consideration* = 2, *little consideration* = 1, and *not applicable* = 0). I did not include zero values in my calculations. For sub-question 3, students did not select 0 (not applicable) as a response for any design feature. I treated all non-responses as system missing information.

4.2 Results

4.2.1 Demographics

The response rate for the demographic questions ranged from 9% to 31%-33% (Table 5). Of the total number of respondents, most were undergraduates (n=43) with 60% in their second or third year of study, while 6.5% identified as master's students.

<i>Table 5 - Percentage of responses to demographic question</i>			
Demographic category	N of responses	N of non-responses	Response rate
Age	13	128	9.2%
Disability	46	95	32.6%
Ethnocultural identity	44	97	31.2%
First generation	46	95	32.6%
Level of study	46	95	32.6%
Gender	45	96	31.9%

Table 5 - Percentage of responses to demographic questions

The majority of the students who responded self-identified as Caucasian or white-European (n=24 out of 44) and as female (n=33 out of 45), which encompassed those who self-identified as female, female, non-binary, and female/woman. Seventy-seven percent of students who provided their age range self-identified as being between 18-23 years old. Eight respondents self-identified as first-generation attendees.

4.2.2 Sub-question 1: Where do you go or avoid on campus when you experience certain emotions?

The first sub-question for the second study asked students *where they would go or avoid depending on the emotions they experience*. Although 141 students responded to the questionnaire, response rates declined steadily throughout the six emotion statements with the highest number of responses being when students indicated “*where do you go when you are happy*” and the least number of responses when students were asked “*where do you avoid when you are disgusted*” (Table 6).

Table 6 – Total number of responses and spaces frequented based on emotion

Statement #	Emotion	Number of responses	
		indicating where students go	indicating where students avoid
1	Happy	141	60
2	Sad	109	54
3	Afraid	88	54
4	Angry	76	51
5	Surprise	63	34
6	Disgust	66	31

Table 6 – Total number of responses and spaces frequented based on emotion

4.2.2.1 Going to spaces on campus

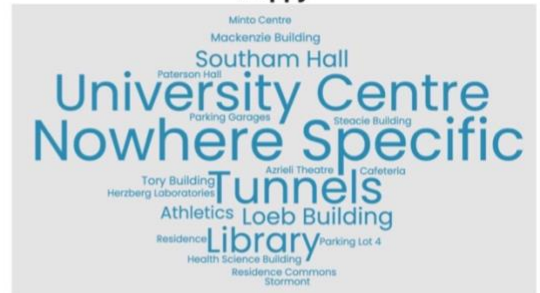
Students named a total of 35 unique campus buildings or designated spaces and two locations not under control of the University (i.e., Rideau River and Rideau Canal) when going to spaces on campus based on certain emotions. The River and Canal are not under the control of the University, and therefore, they are not considered part of the provided built environment. However, I have included them in the results because students frequently mentioned them, suggesting that they may have perceived them as part of their campus built environment experience, especially since both were marked on the map they used for reference. I created

word clouds to help visualize the data (Figure 8). Out of the spaces students approached, seven showed up in reference to all six emotion statements, meaning that students self-reported going to these locations across all emotional states – when happy, sad, afraid, angry, surprised, and disgusted. These spaces were:

1. MacOdrum Library (n=101)
2. University Centre (n=72)
3. Academic Quad (n=26)
4. Richcraft Hall (n=25)
5. Azrieli Pavilion (n=18)
6. Rideau River (n=19) (not under control of the university)
7. Loeb Building (n=10)

Buildings and designated areas students report avoiding when...

Happy



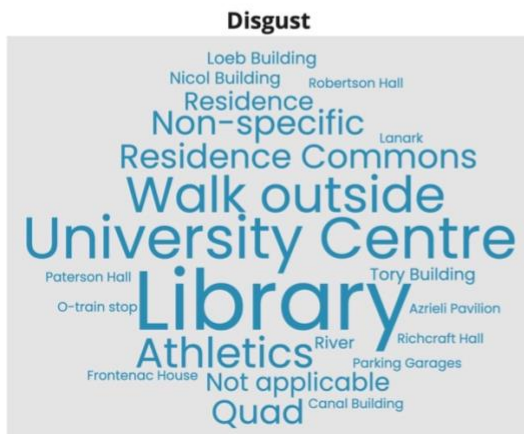
Sad



Afraid



Buildings and designated areas students report going to when...



Buildings and designated areas students report avoiding when...

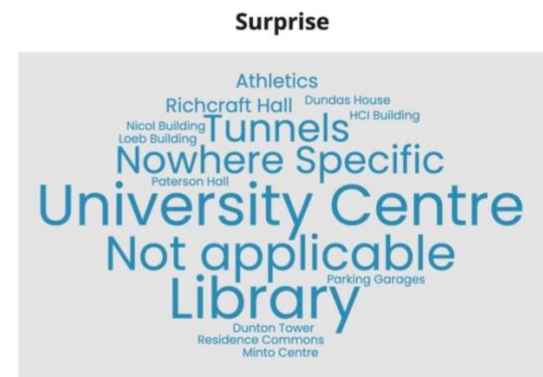
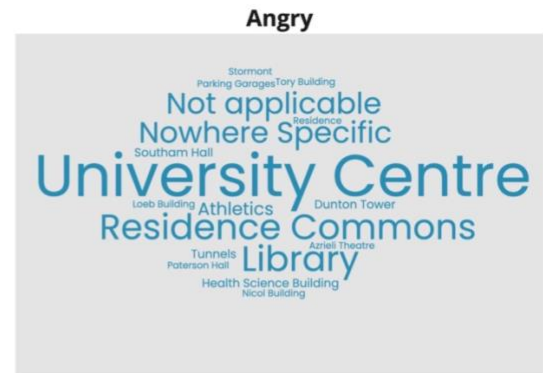


Figure 8 – Word clouds depicting where students reported going or avoiding on campus

The Library was a popular location on campus when going to spaces. Although qualitative data was not specifically asked for, some students provided comments about their experience of the Library. For example, two students referenced seeking out specific spots in the bathroom or the quiet floors when feeling afraid or going there to focus on their work and clear

their mind when feeling *disgusted*. When students were *sad*, comments included using the Library as a “quiet and deserted space” or to access specific floors known as quiet floors (e.g., the third or fifth floors). When *happy*, students self-reported going to the Library to study and do group work, with six students specifically referencing the 4th Floor of the Library, a space that was known for its open, flexible design concept. The 4th floor was also mentioned by a student when experiencing *anger*, citing that they would go there “to vent with friends and distract myself without having to worry about noise.” When experiencing *anger*, another student sought out the Library for reasons related to privacy while a different student expressed that they would go to the Library because it was a calm space where they could work on what was upsetting them. Overall, students reported choosing to go to the Library regardless of their emotions. The selection of this location was especially noticeable when students were feeling emotions that could be negatively perceived such as *sadness* (n=27), *fear* (n=16, tied with the University Centre), *surprise* (n=14), or *disgust* (n=10).

Although the Library was a popular location, there were other places that students frequented depending on their emotions. For example, the University Centre was the most common spot when students experienced *happiness* (n=24) and *fear* (n=16, tied with the Library). One student said when *happy* they would go to the University Centre to have something to eat because it was filled with “positive energy”. As another student described, the University Centre represented a good option to go to when *afraid* given the busyness in the space. Overall, the University Centre was a space that students went to *irrespective of the emotion* they experienced. Athletics and the facilities under their department’s control were also popular spots for students, particularly when they were *angry* (n=17) or *happy* (n=13). However, no specific comments were provided for these choices related to Athletics.

As noted above, some students (n=23) provided additional insights as to why they might go to particular space on campus. Regardless of the space named by students, the most frequented comments across all emotion statements related to sound (n=13) (Table 7). For example, two students wanted to be somewhere they could be comfortable making or being around noise: “4th floor Library to vent with friends and distract myself *without having to worry about noise.*” And “If I want some *background noise*, I will likely go to one of the tunnel seating areas.” Nine students indicated wanting quiet places such as “the *quiet* floor of the Library”, “A *quiet place* in the Richcraft Hall”, or “I will go somewhere for privacy and *quiet* like a study room in the library.” These insights could help inform how designers approach acoustic design in a way that gives options and choice to the users of the space.

Table 7 - Frequency of use of associated words in qualitative comments when going to places					
Direct influences	Number of occurrences	Indirect influences	Number of occurrences	Repetition in text	Number of occurrences
Crowd* or People or Busy* or Alone or Deserted or Priva*	9	Social or People or Commun* Friend*	7	Ugly or Gross	0
Noise or Sound or Quiet or Loud or Traffic	13	Restore or Fatigue or Tired or Stress* Destress*	2		
Light	0				
View or Out*	3				
Air	1				

Table 7 - Frequency of use of associated words in qualitative comments when going to places

The second most often used words related to crowding (n=9). In seven of the responses, the term used showed a desire for space where there were not people. For example, “Tunnels (one of the many seating areas at the tunnel-level of a building where there are *not many other people*)”, or “Try and find somewhere that is *deserted*, with *no one else* (or *very few people* are around) around and is very quiet.” Two students responded that they would seek out specific people as represented by this comment “I may be at Azrieli Pavilion, fourth floor, where other *people of my program* usually are at.”

When I reviewed the data looking at the indirect influences for aspects related to social supports, I noticed more students (n=7) wanting to connect with friends or embrace communal aspects of campus. For example, “If the surprise was good, I would go to places in Richcraft Hall where I would *meet up with friends*.” Another student referenced “The *communal design space* on the 4th floor of Azrieli Pavilion.” These findings suggest that students have different preferences when it comes to the level of crowding as well as use of space for social connection which may need to be factored into design of spaces.

There may have been an interplay between some of the influences as well. For instance, the following statement from a student could be interpreted as having multiple influences. “I feel as though if I were ever afraid on campus, I would go to the University Centre where there is *a lot of foot traffic*. The student could be referring to foot traffic as relevant to direct influences such as *noise* or *crowding*. Yet, the reference could also span both direct and indirect influences, with an overall a sense of security being part of *social supports*. Any of these influences could be relevant individually or collectively which could inform the need for designers to look at how multiple influences could work together to create supportive environments.

4.2.2.2 Avoiding spaces on campus

Among the various campus buildings and designated spaces that students actively avoided based on specific emotions (Figure 8), a total of 27 unique locations were mentioned. Notably, five of these spaces consistently appeared across all six emotional states (happiness, sadness, fear, anger, surprise, and disgust). The following spaces emerged as the most frequently mentioned by students across all emotional states:

1. University Centre (n=54)
2. Library (n=27)
3. Athletics (n=23)
4. Loeb Building (n=15)
5. Paterson Hall (n=6)

The University Centre was at least two times more likely to be avoided by students than any other space. As one respondent stated, “the University Centre crowds worsen my emotion.” Another student indicated that they would avoid the University Centre when *sad* due to noise levels being too loud. More generally, students referenced noise and overcrowding and an overall unappealing environment particularly referencing the associated tunnel system, food court, main indoor pathways, and washrooms. *Regardless of the emotion*, the University Centre was most frequently referenced by students as a space to avoid yet it was most notable when *sad* (n=10), *angry* (n=10), or *disgusted* (n=14). When students were *afraid*, the designated area of Athletics was most frequently avoided (n=8), although students gave no specific reasons for doing so. Students also reported that when *surprised*, they were more likely to avoid the Library (n=6) or the University Centre (n=6) although more generally, two students commented on that it depended if the surprise was good or bad.

When examining Figure 8, an additional pattern emerged, namely that of non-specific spaces. Non-specific spaces referred to spaces that could not be mapped to a specific piece of campus infrastructure or area usually due to the space being referred to in more general terms (e.g., the tunnels). Twenty-seven students indicated that *regardless of the emotion*, they would avoid the tunnels, with two students referring to the tunnel system as “an uncomfortable space” and one student referring to it as one of the “ugly parts of campus”.

Students (n=18) gave additional insights when responding to the question about which spaces they would avoid (Table 8). *Irrespective of emotion*, the most frequent terms reflected the concept of crowds or people (n=10). For example, five students said that they would avoid spaces due to the busyness as reflected in these comments: “I would maybe avoid the tunnels (especially the *busy parts* around the library)”, or “Anywhere that is *busy*.” Another student reflected that it was the crowds in the University Centre because “*crowds worsen emotion*”. Students also commented on indirect influences related to social supports and connections (n=6). For example, students shared comments such as “Patterson, Loeb or busy parts of campus *where my friends would likely be found*,” “avoid the cafeteria because there are *too many people* and germs there,” or “spaces where *my community on campus congregates* (SPPA lounge in Richcraft, near SPPA classrooms).” Designers may want to consider how the proximity of people to each other, levels of activity, sound levels, and privacy in a space can affect people's comfort or discomfort.

Table 8 - Frequency of use of associated words in qualitative comments when avoiding places					
Direct influences	Number of occurrences	Indirect influences	Number of occurrences	Repetition in text	Number of occurrences
Crowd* or People or Busy* or Alone or Deserted or Priva*	10	Social or People or Commun* Friend*	6	Ugly or Gross	6
Noise or Sound or Quiet or Loud or Traffic	5	Restore or Fatigue or Tired or Stress* Destress*	1		
Light	1				
View or Out*	0				
Air	0				

Table 8 - Frequency of use of associated words in qualitative comments when avoiding places

I also saw two words, namely ‘ugly’ and ‘gross’, show up repeatedly in the students’ responses of where they would avoid. Four students stated specifically that they would avoid “the *ugly* parts of campus” with one of those students naming specifically “All the *ugly places* on campus (Loeb, Patterson, Southam, Tunnels).” Two students used the word ‘gross’ to describe places they would avoid. As one of those students stated “I avoid the washrooms at school as much as I can because people are *gross*.” These additional insights reflect the influence of elements such as aesthetics, perception of cleanliness, and orderliness on the space decisions of students.

4.2.3 Results - Sub-question 2: How strongly do coping considerations factor into your decision of where to go on campus?

I asked students to consider particular aspects of coping strategies as reasons they may consider going to spaces on campus. I conducted an analysis on the frequency of responses and the means of the level of consideration given to the coping strategies provided and grouped them according to different emotions (Table 9) which I then visualized as a heat map in Figure 9.

Table 9 - Mean and standard deviations for coping strategies (grouped by emotion statements)

Happy		
Coping strategy	Mean	Standard deviation
Avoidance-based	2.05	0.79
Emotion-based	2.13	0.73
Problem-based	2.00	0.82
Religion-based	1.35	0.62
Social support	2.49	0.71
Sad		
Coping strategy	Mean	Standard deviation
Avoidance-based	2.54	0.76
Emotion-based	2.41	0.76
Problem-based	1.73	0.83
Religion-based	1.34	0.64
Social support	1.68	0.85
Afraid		
Coping strategy	Mean	Standard deviation
Avoidance-based	2.32	0.80
Emotion-based	2.05	0.86
Problem-based	2.07	0.87
Religion-based	1.35	0.71
Social support	1.83	0.91

Angry		
Coping strategy	Mean	Standard deviation
Avoidance-based	2.42	0.77
Emotion-based	2.58	0.74
Problem-based	1.61	0.77
Religion-based	1.36	0.72
Social support	1.79	0.88
Surprise		
Coping strategy	Mean	Standard deviation
Avoidance-based	1.71	0.78
Emotion-based	1.80	0.78
Problem-based	1.98	0.83
Religion-based	1.31	0.64
Social support	2.21	0.83
Disgust		
Coping strategy	Mean	Standard deviation
Avoidance-based	2.50	0.71
Emotion-based	2.50	0.68
Problem-based	2.09	0.86
Religion-based	1.48	0.74
Social support	1.93	0.89

Table 9 - Mean and standard deviations for coping strategies (grouped by emotion statements)

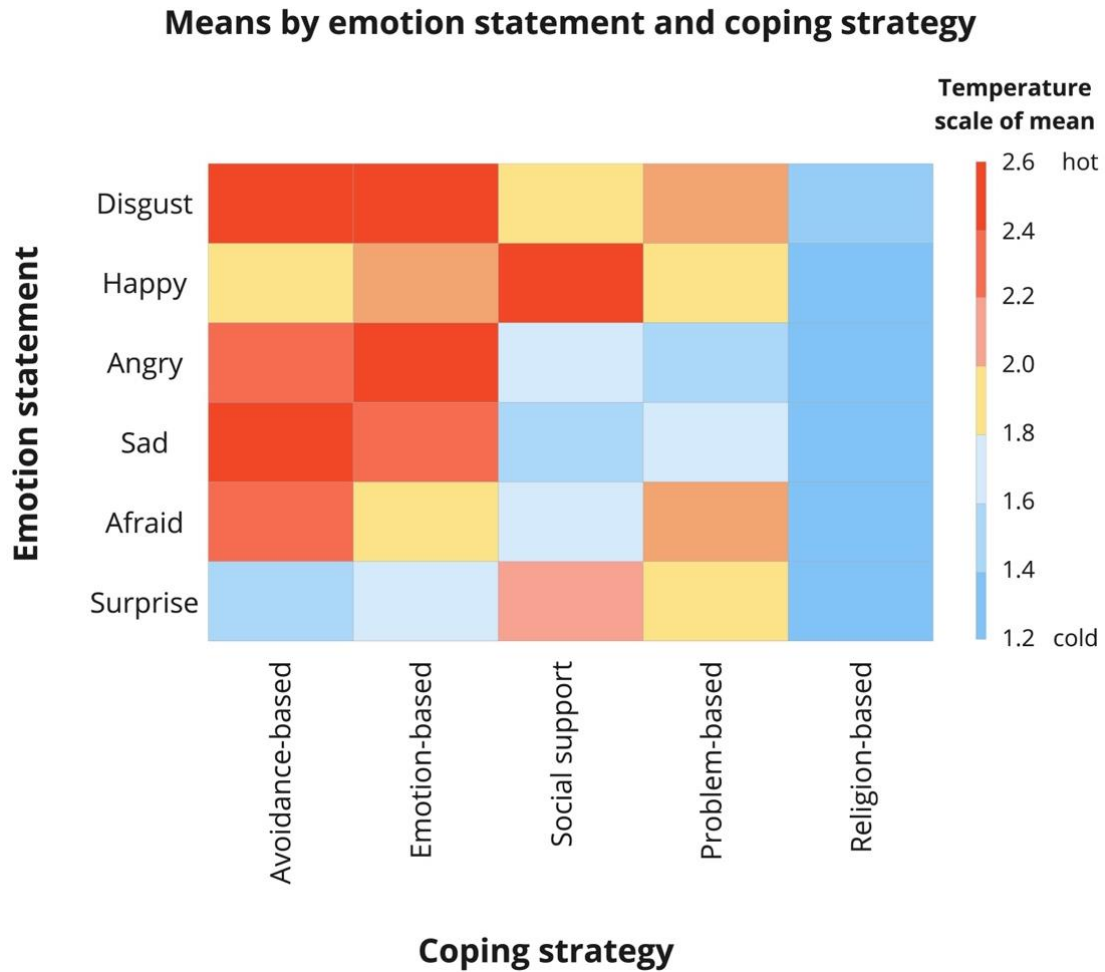


Figure 9 - Heat map of means indicating level of consideration students gave coping strategies in space selection

When students considered coping strategies and experienced perceived negative emotions (e.g., disgust, anger, sadness, and to a lesser extent, fear) they seemed to give stronger consideration to *avoidance-based* and *emotion-based* strategies as part of selecting spaces to go to on campus.

I also noticed a similar level of consideration given to *social supports* when students experienced happiness. The result suggested that designers may want to consider spaces that would support students' desire to avoid stressors, release stress, or promote social engagement. Understanding how individuals' emotions and coping mechanisms influence their choices could help inform the design of spaces to support student mental health and wellbeing.

The coldest spots on the map represented *religion-based* strategy as students were least likely to consider this strategy when selecting a reason for going to a specific space on campus. This result may suggest that religion-based coping strategies as represented in this study were not prominent among students or that students do not consider this strategy in the context of campus. This finding could suggest that designing spaces explicitly catered to religious practices or spirituality may not be a priority for the majority of the student population. This finding could help designers allocate resources and prioritize other aspects of space design that align more closely with the needs and preferences of the users.

Another pattern that I saw emerge in the data is that of *problem-based* strategies, namely that students did not report it as being a strong consideration as to why they went to specific spaces on campus. As a preliminary study looking at student mental health and wellbeing this finding could warrant further exploration as to why students do not have a reaction to spaces in terms of accessing resources and supports to deal with stressors.

4.2.4 Results - Sub-question 3: How strongly do particular design features in a campus space influence your decision to go there?

I calculated the means and standard deviations of the level of influence students perceived specific design features had (Table 10) and then visualized their responses as a heat map (Figure 10). This approach allowed me to see which design features seemed more favoured by students and which were not strongly considered. In looking at the means across all emotion statements, I noted four design features that ranked in the top features of influence when students were selecting space, that is to say consistently showed as ‘hot spots’ on the map. Participants perceived *layout*, *orderly and clean*, *access to natural light*, and *noise control* as having a

stronger influence across all emotion statements which could indicate these features are prominent in space preferences of students.

Table 10 - Mean and standard deviations for design features grouped by emotions (in alphabetical order)

Happy			
Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	88.00	69.09	31.43
Access to comfortable furniture	87.00	65.17	26.67
Access to natural lighting	90.00	76.00	25.30
Connect meaningfully with others	88.00	64.66	29.32
Flexible, adaptable, accessible	91.00	62.75	27.29
Fosters sense of community	89.00	65.06	30.46
Indoor air quality	75.00	56.27	27.84
Layout	97.00	65.26	25.25
Noise control	83.00	54.34	31.94
Orderly and clean	96.00	69.79	26.99
Quality of indoor lighting	82.00	63.54	25.98
Supports diverse cultural practices	48.00	32.29	32.70
Temperature control	85.00	57.88	25.73
Sad			
Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	73.00	65.07	32.02
Access to comfortable furniture	75.00	66.00	30.58
Design Features	Responses	Mean	Standard Deviation
Access to natural lighting	74.00	65.54	32.27
Connect meaningfully with others	61.00	46.07	32.78
Flexible, adaptable, accessible	65.00	52.77	31.05
Fosters sense of community	62.00	48.39	33.15
Indoor air quality	62.00	50.65	29.08
Layout	79.00	73.42	24.12
Noise control	77.00	74.29	26.68
Orderly and clean	76.00	70.53	26.53
Quality of indoor lighting	69.00	58.12	28.92
Supports diverse cultural practices	24.00	32.08	32.43
Temperature control	73.00	56.44	28.98
Afraid			
Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	61.00	56.39	35.31
Access to comfortable furniture	70.00	60.29	31.85
Access to natural lighting	61.00	62.13	32.61
Connect meaningfully with others	60.00	51.83	36.34

Design Features	Responses	Mean	Standard Deviation
Flexible, adaptable, accessible	59.00	56.44	29.81
Fosters sense of community	63.00	58.73	34.62
Indoor air quality	57.00	54.74	30.54
Layout	68.00	70.88	25.49
Noise control	73.00	64.11	36.09
Orderly and clean	74.00	62.70	31.81
Quality of indoor lighting	64.00	56.88	30.02
Supports diverse cultural practices	21.00	32.38	32.08
Temperature control	67.00	52.39	30.36

Angry

Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	57.00	59.82	33.57
Access to comfortable furniture	50.00	52.60	33.25
Access to natural lighting	57.00	65.79	29.70
Connect meaningfully with others	51.00	50.20	31.21
Flexible, adaptable, accessible	53.00	52.45	26.30
Fosters sense of community	55.00	51.09	31.90
Indoor air quality	42.00	50.24	27.72
Layout	63.00	69.37	23.89
Noise control	60.00	58.50	31.56
Orderly and clean	59.00	67.12	29.07
Quality of indoor lighting	48.00	57.92	27.52
Supports diverse cultural practices	16.00	36.88	33.01
Temperature control	56.00	55.71	27.16

Surprise

Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	47.00	57.66	31.43
Access to comfortable furniture	55.00	68.00	25.05
Access to natural lighting	48.00	61.04	30.05
Connect meaningfully with others	52.00	65.96	30.70
Flexible, adaptable, accessible	43.00	57.44	29.28
Fosters sense of community	52.00	64.42	30.12
Indoor air quality	43.00	50.93	26.17
Layout	56.00	64.46	26.90
Noise control	56.00	60.00	30.81
Orderly and clean	57.00	66.67	26.68
Quality of indoor lighting	51.00	57.06	26.33
Supports diverse cultural practices	14.00	31.43	30.60
Temperature control	48.00	53.13	27.69

Disgust			
Design Features	Responses	Mean	Standard Deviation
Access to a view of nature	39.00	56.92	34.12
Access to comfortable furniture	40.00	55.75	32.02
Access to natural lighting	43.00	61.63	32.73
Connect meaningfully with others	39.00	53.59	29.60
Flexible, adaptable, accessible	36.00	59.17	30.18
Fosters sense of community	39.00	54.87	29.99
Indoor air quality	33.00	47.58	29.37
Layout	45.00	67.11	26.43
Noise control	46.00	63.04	27.88
Orderly and clean	44.00	67.27	25.91
Quality of indoor lighting	37.00	52.16	29.07
Supports diverse cultural practices	11.00	33.64	33.55
Temperature control	41.00	50.00	29.24

Table 10 - Mean and standard deviations for design features grouped by emotions (in alphabetical order)

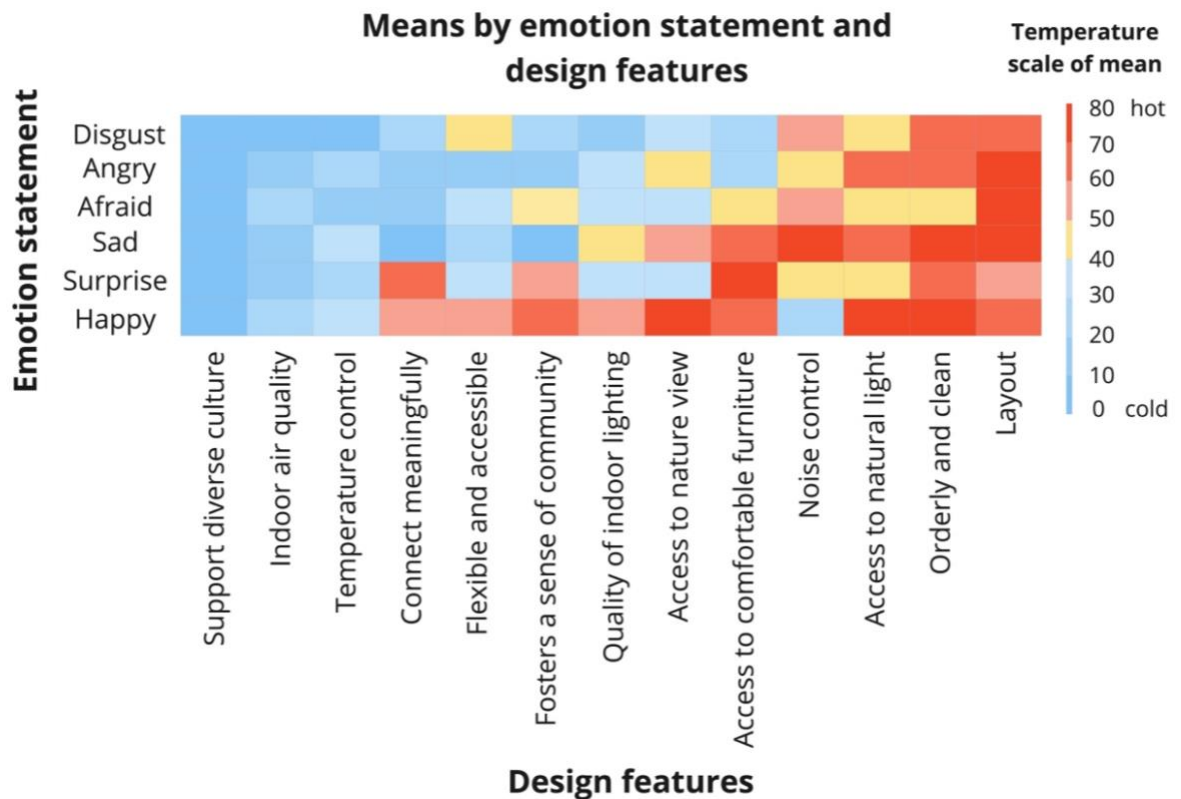


Figure 10 - Heat map of means indicating level of consideration students gave design features in space selection

When looking at the ‘cool spots’ across all emotion statements, those with perceived less influence were *space that supports diverse cultural practices*, *indoor air quality*, *temperature control*, and *connect meaningfully*. This could indicate that students give less consideration to these aspects of design in campus spaces. That said, *connect meaningfully* appeared to have the largest difference in terms of means, ranging from 46 when students were sad to a high of 66 when surprised. This could indicate that while not a popular consideration for space overall, when students are seeking more meaningful connection, it is a warranted consideration in certain instances (e.g., when surprised or happy).

Additional patterns emerged when looking at emotions that could be perceived as negative, namely sadness, anger, surprise or disgust. When students experienced these emotions, they were more likely to consider features such as *layout* and *orderly and clean*. Students experiencing negative emotions were less likely to consider space because of *indoor air quality* and *temperature control*. When it came to *noise control*, there was a slightly different pattern. This feature was likely to be given stronger consideration when experiencing sadness, fear, and disgust which could suggest that noise control is more of a design priority when students experience those feelings.

Lastly, the majority of the features had a mean score above 50.00 regardless of the emotion experienced, meaning that students perceived them as having some influence in their consideration of space. The most notable exception to this finding was with *support diverse cultural practices* where the mean was at its highest point when students were angry (M=36.88, SD=33.01) and lowest when surprised (M=31.43, SD=30.60). Three other features had specific instances where the mean scores dropped below 50.00. *Indoor air quality* did not feature in the top half of features when students experienced *disgust* (M=47.58, SD=29.37).

4.3 Summary

In the first study of the second research question, I asked the students to *identify places on campus, based on their emotions, that they would either visit or avoid*. The students identified a total of thirty-five unique campus buildings or designated spaces, and two locations that were not under the University's control, namely the Rideau Canal and the Rideau River. Among the identified locations, the Library was a popular spot, and students reported choosing to visit it *irrespective of their emotions*. The Library attracted students to its space for studying, group work, and reflection. On the other hand, students avoided the University Centre and non-specific spaces like the tunnels *regardless of their emotions* due to noise, crowds, and the unappealing environment.

My objective of the second sub-question was *to explore the reasons why students choose certain campus spaces based on their coping strategies*. The findings suggested an inclination by students experiencing *negative emotions* to consider *avoidance-based* and *emotion-based* coping strategies when selecting spaces on campus. Furthermore, students did not seem to prominently consider space selection to support *religion-based* coping strategies. Overall, understanding how students' emotions and coping mechanisms influence their space preferences can inform the design of spaces that support mental health and wellbeing. Designers can focus on creating environments that facilitate avoidance of stressors, provide opportunities for stress release, and promote social engagement. However, the study suggests that designing spaces explicitly catering to religious practices may not be a priority for most students.

Additionally, the study explored the *design features* that influenced students' selection of campus spaces across different emotions. The *layout, access to natural light, orderliness and cleanliness*, and *noise control* were the top design features that influenced students *irrespective*

of the emotion. On the other hand, *space that supports diverse cultural practices, indoor air quality, temperature control, and connect meaningfully* were less influential. However, the majority of the design features still had some influence on students' consideration of space. By understanding these patterns and preferences, designers can make informed decisions about creating spaces that effectively meet the needs of students. It allows designers to consider the influential design features and the impact of emotions on students' perceptions of space. This knowledge can support to the creation of more engaging, functional, and inclusive environments.

Chapter 5: Discussion

5.1 Research question one

My first research question asks if existing literature indicates a relationship *between the post-secondary campus built environment and students' mental health and wellbeing*. The results of the research point to a possible relationship across many aspects of campus including learning environments, dormitories, and outdoor landscapes. Although sparse, the literature in the rapid scoping review suggests there may be an interplay between the built environment and student emotional processing, mood, reaction to stress, and cognitive function.

By analyzing the results of the rapid scoping review using the concepts of general resistance resources (GRRs) and specific resistance resources (SRRs), part of the salutogenic framework, I hoped to gain a deeper understanding of how students related to the campus built environment. Conducting research that considers both types of resources is foundational to the concept of salutogenesis as it allows us to probe the personal characteristics and social supports of students in the context of environmental factors. According to Mittlemark (2017), our health and the ease/dis-ease at which we navigate it is not by chance; it is reflected in our sense of coherence, which is informed by GRRs and SRRs. Thus, ultimately, how GRRs and SRRs interact impacts our overall health and wellbeing. Yet, in using the salutogenic resource categories as a coding method for the results, only two of the articles looked at both GRRs and SRRs (Marchand et al. and Peters & D'Penna).

In his early work, Antonovsky advocated for a need to study a person's coping (GRR) and the provided environment (SRR), because collectively GRRs and SRRs affect our sense of coherence (Antonovsky, 1996; Mittlemark et al., 2017). Salutogenesis brings forward the notion

that there is an interplay occurring between the internal and external resources available to someone. By looking at GRRs and SRRs as independent or in isolation from each other, there is a risk in outcomes of research missing the nuances of the interplay and the subsequent impact that can have on a person's sense of ease/dis-ease. This segmentation is contrary to a sense of coherence which calls us to embrace the integration of the person and their environment as an open system (Vinje & Langeland, 2017), that is to say, the acceptance of a symbiotic, fluid relationship with the environments around us.

A similar pattern repeats itself when I examined the results from coding the articles based on direct and indirect influences of the built environment, as defined in Evans (2003). One of the studies (Marchand et al., 2014) looked at both direct and indirect influences. Like GRRs and SRRs, there is an interrelatedness between direct environmental factors such as light, sound, housing quality and internal resources (e.g., social support, personal control) we have available to us at any given time. Marchand et al. (2014) focused on lighting, sound, and temperature controls (i.e., direct influence), and how these direct factors may be a consideration in students' emotions and ability to learn, including effects such as cognitive fatigue (e.g., indirect influence). We see this same interrelatedness in a research study referenced in Evans & Stecker's (2004) systematic review which identified a relationship between students living in noisy, crowded dormitories and learned helplessness.

The results of the first research question highlight the importance of looking at post-secondary students' mental health and the built environment. While it is important to scope research accordingly by looking at aspects of GRRs and SRRs, if we isolate them entirely from each other we could miss out on new understandings regarding the relationship. Additionally, by polarizing resistance resource categories that may be within a student (e.g., GRRs represented as

emotions and coping) versus what resources might be out of their immediate control (e.g., the campus built environment as an SRR), students could experience decreased mental health outcomes due to the burden of systemic barriers.

The findings of the rapid scoping review demonstrated that even within the small body of relevant research, there was not a focus on understanding the context of the built environment as students might experience it. Previous research seems to approach campus space as more of a utility with a practical and functional component (e.g., a student goes to classrooms to learn, the library to study, or a café to eat). Yet, perhaps students experience the campus in a multitude of ways that go beyond this functional approach and in a way that could indicate space as having multiple uses and contexts (e.g., a student goes to a classroom to escape their day, the library to be alone, and the café to unwind with friends). The second study of my research represents an opportunity to close this gap by asking students to share their experiences regarding the campus built environment as part of their mental health and wellbeing.

5.2 Research question two

The second broad research question explores *the relationship between coping strategies and design features on campus and students' considerations of where to go when experiencing different emotions*. As students rely on campus infrastructure provided by the institution while engaging in a breadth of activities (e.g., housing, nutrition, exercise, socialization), I wanted to gain insight as to what this reliance may look like in light of their overall mental health and wellbeing. I broke the study down into three sub-questions in order to address the primary question.

5.2.1 Emotion and space selection

I aimed to capture personal experiences of students regardless of specific stressors or specific environments that they may experience as part of student life by focusing on emotion statements. The rationale for this stems from the notion that emotions act as a personal barometer that can indicate how we may respond to particular stressors (Zautra et al., 2005). Furthermore, the emotional experiences of students may shape how they create meaning with spaces. This involves a shift from mere spaces to the notion of ‘place,’ encompassing the physical, cultural, and aesthetic elements that shape people's individual and social engagement (Banning et al., 2010; Huggins, 2022; McLane & Kozinets, 2019). While ‘place and space’ were outside the scope of my research, the results of my research suggest that students' emotional experiences may influence their choice of spaces. The identification of specific locations that students frequent when experiencing particular emotions suggests that the built environment plays a role in shaping student experiences on campus.

While students indicated spaces in the Library that they would go to more than any other campus spaces, they also frequented the University Centre. Conversely, students avoided more spaces in the University Centre than other parts of campus but named the Library the second most frequented space they would avoid. Why are students avoiding and seeking out the same spaces irrespective of emotion? There are several factors that contribute to these locations as attractors and detractors. First, we may need to understand a bit more about the space at libraries and university centres (e.g., student centres) at post-secondary institutions to understand the reasons why the space can attract and detract across emotional experiences.

Based on my research, spaces like student centres (e.g., University Centre) and libraries are an important consideration in students’ decision-making. Students use the spaces, in part, to

support their overall experiences. Having spaces where students can discuss, debate, and converse are environments where “significant learning can occur” (Kolb & Kolb, 2005, p. 208) while fostering connection with peers. Campus libraries and student centres may offer resources such as study spaces or access to technology, which can be valuable assets to students. This is representative in the evolution of informal learning spaces such as library cafes becoming multi-functional hubs to support student experiences (Deng et al., 2019). Other research studies have found that student hubs and libraries provide opportunities for flexible, more personalized experiences particularly around socialization, connection, and community building, which can positively impact their overall post-secondary experience and academic success (Deng et al., 2019; Hurst, 2013; Lugosi, 2019; Mclane & Kozinets, 2019). Qualitative data from my research supports those studies in that 31% of the students who provided more detailed responses sought spaces that would support the level of social engagement or connection they might be seeking. For instance, one student would go to the “4th Floor Discovery Centre to vent with my friends” while another would go to their student society “where many of my friends spend time”.

Social connection can act as an attractor to space, reflected in sentiments of students (n=7) that referred to communal spaces, wanting to meet up with friends, or being with other members of their program. Previous research on mental health and wellbeing and the built environment echoes the desire for the built environment to support social needs and connection (Deng et al., 2019; Hurst, 2013; Lugosi, 2019; Mclane & Kozinets, 2019). However, it is important to note that social connection can also act as a deterrent in certain spaces. This is supported by feedback from students (n=6) who expressed that some areas felt crowded due to an excessive number of people, or they actively avoided certain spaces if their friends were present, preferring solitude instead.

The qualitative data from this study brings in a different layer of insight about emotions and selection of spaces on campus. For instance, depending on emotions, sound played a role more often as a detractor than an attractor. Of the students who provided more insights into their respective space selection based on the emotion statements, 44% used words I grouped as indicating sound is a detractor, whether going to or avoiding spaces on campus (Table 7, Table 8). Furthermore, 44% indicated words that would imply crowds or busyness of space were detractors as well. Collectively, these findings could indicate that influences such as noise and crowds factor into students' space decision making when going to or avoiding spaces. This finding provides support for other research that demonstrates environmental factors such as noise, overcrowding, and lack of privacy which can negatively affect mental health and academic performance (Evans & Kim, 2013; Kuo & Sullivan, 2001). Sullivan & Chang (2011) noted that prolonged exposure to crowded and noisy environments can result in poor psychological outcomes including depression.

The concept of crowding extends beyond the physical space itself. Our social, cultural, and psychological contexts shape our spatial behavior and influence the interpersonal distances we feel comfortable with (Rios-Martinez et al., 2015). Consequently, these factors can impact our level of comfort and shape our social interactions (Nasrallah & Pati, 2021). The notion of comfort/discomfort is captured in the theory of proxemics, namely the study of the "spatial distances individuals prefer to maintain in various social and interpersonal situations" (Rios-Martinez et al., 2015, p.139). As a result, students may actively avoid crowded and noisy areas on campus, seeking solitude, comfort, and stress reduction.

My findings also suggest that restorative spaces on campus may play a role in how students use space as part of coping. Students sought restorative spaces although the majority of

the spaces named are not in control of the University. This is most notable when students referenced the Rideau River which showed up across all emotion statements. That being said, four students referenced the indoor space of a building on campus, Richcraft Hall, which has windows that overlook the River as well as a living plant wall with a water feature. Restoration spaces often promote renewal of cognitive energy (i.e., attention restoration), emotional management (i.e., emotional regulation) and stress management, such as quiet spaces for reflection and relaxation (Evans, 2003c; Hoisington et al., 2019; Peters & D’Penna, 2020; Shepley et al., 2016; Wyles et al., 2019). Research has shown that various types of landscapes and environments, including green spaces and areas with pleasant and natural sounds, can have a significant impact on the restoration of psychological and physiological function (Hipp et al., 2016; Kaplan & Berman, 2010; Peters & D’Penna, 2020; Zhang et al., 2023).

5.3 Coping and space selection

Looking at space selection based on emotion illuminates one motivation of student decision-making regarding the campus built environment. The second sub-question explores how students may use the built environment as a resource to support coping strategies. Previous research demonstrates a relationship between environmental stressors in the built environment and our ability to cope (Codinhoto et al., 2009; Evans, 2003; Sullivan & Chang, 2011), although none of these studies involved students at post-secondary institutions. I focus on coping strategies as one characteristic of mental health and wellbeing as they are an indicator of how we assess and respond to stressors (Anisman, 2016).

In my study, I found that students make space selection relative to avoidance-based and emotion-based strategies, particularly when experiencing negative emotions. For example, students indicated that they were more likely to seek out campus spaces as an avoidance-based

strategy when experiencing sadness. Similarly, when students experience anger, many of them report going to Athletics to release emotion or let off steam (i.e., emotion-based strategy). Aspects of previous research supports similar findings. For instance, in Ben-Zur & Zeidner (2012) students who perceived academic stressors as more stressful tended to rely more on emotion/support coping and avoidance coping strategies. A study in Botswana noted that students preferred emotion-based and problem-avoidance strategies (Monteiro et al., 2014).

In the heat map (Figure 9), I noticed that students did not exhibit a particularly strong consideration for problem-based strategies as part of their space selection. This is not to say that students did not give consideration to problem-based strategies, but more that they were often not the first or second most considered strategies when deciding where to go on campus. Literature links problem-based strategies with concepts like resilience, enhanced feelings of control, and a reduction in stress effects (Anisman, 2014; Freire et al., 2020; Saleh Baqutayan, 2015). Part of my findings may be explained by the age group of the respondents. Barquín-Cuervo et al. (2018) and Ramos et al. (2021) (as seen in Raquel Suriá Martínez and Juan Manuel Ortigosa Quiles, 2022) and (Taleb et al., 2020) propose that during adolescence and youth, individuals tend to seek more support from their peer group rather than employ problem-based strategies. While studying the use of coping strategies is outside of the scope of my research, these findings do raise questions about why problem-based strategies are not reported as influential in accessing spaces on campus. The results may also indicate a co-employing of strategies, for instance a student could be combining emotion-based strategies and problem-based strategies while seeking out resolution to a particular stressor.

Another pattern repeated consistently across emotion statements, namely that of religion-based strategies (represented in this project as “spiritual needs”). On the heat map (Figure 9), this

strategy consistently was in the ‘cool’ zone, meaning it was a lower considered strategy across all emotions. While this could be as a result of declining religious beliefs in students (Torralba et al., 2021; Downey, 2017), this can also be because students do not identify campus space as being part of their expression of religious beliefs or practices.

Overall, the study gives us early insights as to how students' emotions and coping strategies influence their selection of spaces on campus and their overall mental health and wellbeing. The findings suggest that students tend to rely on avoidance-based and emotion-based coping strategies, particularly when experiencing negative emotions. Problem-based strategies are less commonly considered while religion-based strategies a rarely a strong consideration. This study represents a first step and perhaps future research can provide more robust attention to the intersection of these topics. By understanding how students use and respond to different spaces on campus, administrators and planners can create more supportive environments that promote wellbeing.

5.3.1 Design features and space selection

The third sub-question builds on the previous two sub-questions by probing how students may rate specific features in the space they originally commented on. Four of the five top-ranked design features that students perceive as having the most influence on their selection of campus spaces include layout, orderly and clean spaces, access to natural light, and noise control. Collectively, layout, noise, and orderliness features are sub-characteristics of direct influences of the built environment while access to natural light spans both direct (e.g., light) and indirect (e.g., restoration) (Evans, 2003). These features may also inform other aspects of mental health and wellbeing including comfort, concentration, and overall satisfaction in the environment (Vischer & Wifi, 2015; Shepley & Pasha, 2017). This is of particular importance in post-

secondary environments where a student's perception of the built environment can impact motivation, concentration and comfort (Peters & D'Penna, 2020; Yildirim et al., 2011), all of which contribute to the learning process.

5.3.1.1 Layout

In the rapid scoping review I conducted, Licite & Janmere (2018) analyzed formal and informal learning environments in different academic disciplines at a university in Latvia. Among the elements evaluated by the team, layout and room size accounted for two of the nine factors students were asked to rate. The layout of rooms received the highest rating among the physical environment aspects ($M=4.67$, $SD\ 1.76$) (Licite & Janmere, 2018). Similarly, in my own study, the layout was identified as the most significant feature across various emotions, with mean scores ranging from 64 (when surprised) to 73 (when sad) out of 100 (Figure 10).

The Latvian study also found that students showed a preference for informal learning environments (58%) (Licite & Janmere, 2018). In my study, the Library and the University Centre emerged as the most frequently visited and avoided spaces. Both of these locations provide informal areas that support student learning (Kolb & Kolb, 2005). Interestingly, in qualitative comments, students mentioned going to formal learning spaces (e.g., classrooms, studios, auditoriums) only when expressing sadness, as exemplified by statements such as, 'I would go to a small room in AP 4th floor, empty classroom, or empty studio space in AP 4th floor.' On the other hand, when avoiding spaces, students explicitly stated their avoidance of classrooms when happy or sad. As one student explained, 'I would avoid any classroom buildings with lecture halls because they stress me out.'

5.3.1.2 Orderliness

Orderliness is a component of salutogenic design, represented in the comprehensibility of the environment, in other words, when an environment is predictable, consistent, and explainable

(Eriksson & Mittelmark, 2017). In Shepley et al. (2016), the team conducted a study looking at the design of the built environment for mental health facilities. While there are fundamental differences between a post-secondary environment and mental health facilities, both are complex organizations that are public facing with their own sense of identity and community, thus the results of the research are interesting considering the absence of studies at post-secondary institutions on this topic.

Interestingly, in Shepley's study, participants ranked the concept of an "orderly and organized environment" as the least important design characteristic, although it still received a relatively high mean score of 5.8 out of a possible seven. This finding contrasts somewhat with my own research, where the combination of an *orderly and clean* environment emerged as the second most popular feature consideration across all emotion statements with means ranging from 63 (when afraid) to 71 (when sad) out of a possible one hundred (Figure 10). It is important to acknowledge that I grouped "orderly and clean" together as one feature, while Shepley et al. separated out the variables representing maintenance of the environment from the concept of order or organization. Therefore, it remains unclear in my study whether there is a preference for orderly environments versus clean environments.

The perception of order and cleanliness in an environment has subjective components. However, it is interesting to note that there can be a certain tolerance for complexity in a space that invites people in. Shepley et al.'s study captured this impact of complexity well, as one respondent stated, *"The complexity and a little bit of lack of order actually makes most members... [more comfortable because] it allows them more autonomy and activity than something that's very prescriptive and orderly"* (Shepley et al., 2016, p. 50). This quote exemplifies the need for comprehensibility to the extent that it makes cognitive sense and allows

for a sense of agency among users of the built environment (Golembiewski, 2022). Future research may want to look at the comprehensibility of design features such as orderliness or cleanliness and how they factor into a student's desire to predict the campus built environment.

5.3.1.3 Access to natural light

While students placed strong emphasis on natural light across all emotions, the fact that it was not rated as highly important during negative emotions like sadness surprised me. Natural light significantly impacts both physical and mental health (Brennan et al., 2002; DuBose et al., 2018; Hoisington et al., 2019; Vischer & Wifi, 2015). Studies conducted in workspaces have shown that natural light can enhance motivation and improve mood (Hoisington et al., 2019; Vischer & Wifi, 2015), whereas a lack of natural light can act as a physical stressor (Brennan et al., 2002; Hoisington et al., 2019). In healthcare settings, access to daylight has been associated with reduced pain, decreased depression, and improved sleep (DuBose et al., 2018). Given these findings, I did not expect that students would consider natural light more strongly when happy.

Improving access to natural light or integrating natural elements and nature-replicating elements into the built environment, may help in students' emotional regulation. Natural elements support restorative environments, thus enhancing students' access to their general resistance resources, such as coping and mood regulation. As an example, dynamic and diffuse lighting, which utilizes varying intensities of light and shadow to simulate natural conditions, is a key element of biophilic design. Previous research notes this type of lighting, along with a connection to the natural world (e.g., windows offering views of temporal and seasonal changes), can reduce stress and improve emotions, mood, and preferences (Browning et al., 2014; Peters & D'Penna, 2020). Design elements that evoke nature and incorporate sensory variability can enhance concentration (Browning et al., 2014) and lead to improved performance (Peters & D'Penna, 2020), both of which are essential for student success.

5.3.1.4 Noise control

Noise control emerged as the fourth most considered design feature across all emotions, with mean scores ranging from 59 (when angry) to 74 (when sad). The qualitative feedback provided by students in response to sub-question one supported this finding, with 18 students specifically mentioning noise as a detractor in spaces on campus. Additionally, their comments about crowds and busyness indicated that acoustical comfort could also be influenced by these factors.

The importance of noise control for mental health and wellbeing is a critical consideration. Previous studies have shown that even noise annoyance, such as being irritated by noise rather than its decibel level, can have negative impacts on mental health outcomes, particularly in lower quality housing or high-density residential units (Evans, 2003c; Hoisington et al., 2019). Noise annoyance has been linked to poorer mental health, chronic stress, (Codinhoto et al., 2009; Hoisington et al., 2019; Sullivan & Chang, 2011), elevated levels of cortisol (Recio et al., 2016), and academic performance (Felsten, 2009; Onchang & Hawker, 2018; Shield & Dockrell, 2008).

In the context of salutogenic design, noise control is considered an essential element that supports wellness. Noise interferes with our ability to manage and make sense of stressors, both physiologically and psychologically (Browning et al., 2014), thus affecting our overall sense of coherence. Similar to my study, Shepley et al. (2016) also ranked noise control as an important feature in their research. They further explored the effectiveness of the built environment in facilitating noise control, finding it to be the least effective feature. Although I did not measure the effectiveness of the features in my study, this presents an important opportunity for future research.

5.3.1.5 Meaningful connection and support for diverse cultural practices

Meaningful connection and support for diverse cultural practices may be perceived as less influential in students' decision-making because they have a more indirect impact on their immediate experience. In my study, meaningful connection ranked in the bottom four features and support for diverse cultural practices ranked last. While these findings may indicate that these features are not a priority for the majority of students, it is crucial to recognize that these features should not be disregarded. For instance, support for diverse cultural practices may not be a primary concern for students who do not identify with specific ethnocultural groups but for those who identify in those groups, the consideration for this feature could be higher.

Combined, connect meaningfully and support diverse culture play a vital role in creating a welcoming campus environment. For instance, campus spaces that accommodate diverse cultural practices can foster a sense of belonging (CSA, 2020; Museus et al., 2018) and encourage cultural awareness and understanding and as such are an important psychosocial factor for mental health and wellbeing (CSA, 2020). Other research demonstrates that institutional supports (i.e., specific resistance resources) are critical components that influence a students' sense of belonging and as such, policies, structures, and practices are needed to reduce feelings of othering (Museus et al., 2018; Owusu-Agyeman, 2021). Overall, salutogenic design approaches can foster a sense of belonging which in turn impacts students' levels of enjoyment, motivation, and achievement (Pedler et al., 2022).

5.4 Contributions to the field

More broadly, my research attempts to respond to the three gaps identified from the salutogenesis research community. Throughout his career, Antonovsky expressed criticism regarding the emphasis researchers placed on the sense of coherence as an independent variable,

causing ease or dis-ease, rather than recognizing it as a dependent variable (Mittlemark et al., 2017). I attempt to address this by looking at aspects of generalized resistance resources (GRRs) and specific resistance resources (SRRs) as dependent variables (GRRs being coping strategies and SRRs being spaces and design features) that could potentially influence the sense of coherence in students. Secondly, Vinje et al. (2017) critique that there is a notable body of literature that predominantly examines generalized resistance resources (GRRs) while neglecting the examination of specific resistance resources (SRRs) alongside them. My research responds to this criticism by looking at the campus built environment as an SRR and the relationship it has in the coping strategies of students (i.e., GRR). By anchoring my research in this way, I respond to Mittlemark et al. (2017) who posit that there is a scarcity of research exploring if it is plausible that SRRs (i.e., campus built environment) play a role in a person's sense of coherence. While evaluating a sense of coherence was outside the scope of my study, my research begins to explore the concept of the campus built environment as an SRR and how it may play a role in students' coping strategies.

Given the scarcity of research looking at the built environment and student mental health, it was not surprising that there wasn't a comprehensive list of design features that could support or hinder students' perception of mental health and wellbeing. While there is a standard that provides guidance on post-secondary student mental health (CSA Z2003:20), the role of the built environment in student mental health is sparse. To break down design features representing aspects of the built environment, I thoroughly reviewed studies that examined the relationship between the built environment and mental health on a broader scale. This scan spanned work in mental health facilities, healthcare institutions, and workplaces. Based on that research, I created a matrix of design features that may support the mental wellbeing of students (Table 1).

However, further research is necessary in this area to guide the development of evidence-based resources to inform design development, and evaluation techniques such as space audits.

The insights gained from this project may be of interest to various stakeholders in light of mental health challenges being on the rise. Facilities and space planning teams may want to consider the use of the matrix developed in this study to inform design principles related to supporting student mental health and wellbeing. Design teams may also benefit from learning more about how particular spaces are used or viewed by students as part of their mental wellbeing. Student mental health promotion teams may find it useful to bring their programming to where the students are going when experiencing more negative emotions to aid in access to supportive resources.

5.5 Limitations

5.5.1 Campus built environment during the COVID-19 pandemic

My research took place during the COVID-19 pandemic, specifically March-April 2022. The measures implemented to control the spread of the virus, such as social distancing and remote learning, resulted in substantial changes to the campus environment. By Spring 2022, while there was a partial return to classes on campus, the shift to remote forms of learning altered the physical experience of being on campus for students including the use of and access to facilities such as residences, libraries, study spaces, and recreational areas, and dining venues. These changes may have disrupted the sense of community and connection that the campus environment typically provides, which could contribute to feelings of isolation and loneliness among students (Sillcox, 2022), influencing the results of the research. Additionally, the pandemic introduced new stressors, such as online learning, that have affected student mental health. Fatigue, loss of motivation, isolation and loneliness, depression, anxiety, and post-

traumatic stress disorder impact today's post-secondary students (Sillcox, 2022; Impacts of COVID-19 on Canadian Post-Secondary Education, 2023; The Impact of COVID-19 on Post-Secondary Institutions, n.d.).

5.5.2 Study Design

From the outset of the project, the intent was to gather and analyze data that would provide insights into the context of campus spaces in relation to student mental health and wellbeing. While the research approach was reflective of this, in hindsight, the complexity of statistically analyzing the data presented a challenge. I chose to focus on descriptive and qualitative analysis for the preliminary analysis of the dataset in this exploratory research. This approach helped me gain a deeper understanding of the participants and laid the foundation for future statistical analyses (Fulk, 2023). Further statistical analysis at this time would have required significant investment on my part. These decisions resulted in not directly assessing the relationships among responses to the various constructs, and hence not being able to answer the overarching research question, namely, *what is the **relationship** between the post-secondary campus built environment and students' mental health and wellbeing*. Future studies may benefit from having a diverse team of researchers, including those who can complete inferential statistics to determine if there are any conclusive findings from the data (APA Dictionary of Psychology, n.d.).

An additional limitation in the design of the study was how I presented the question relating to design features to students. The wording of the question, "*What about this particular space appeals to you?*" presents design features independently from the previous two questions which explore *where students go* and *why they may go there* (based on coping strategies). The phrasing of the question prevented me from observing patterns between coping strategies and

design features and without inferential statistics, it was not possible to probe possible relationships between these variables based on the statistics I chose to present.

How many students interpreted the one example or representation they were given for each coping strategy (e.g., social supports represented as hanging out with friends) could have been misleading or too restrictive a representation of the strategy. Future studies may want to provide multiple examples or representations for each coping strategy or offer a range of scenarios. This idea of interpretation of what a term could imply could arise with design features as well (e.g., layout, connect meaningfully, support diverse cultural practice) in that the term may be too vague and or open to vast interpretation which could influence the findings. Future studies could provide specific definitions to help with clarity of the features.

I also made an inherent assumption that the students would have a certain level of self-awareness regarding their emotional states and that they could make the connection between that state and selection of space on campus. For this study, I provided little context or definitions to the participants. I asked students to use their best judgement in defining what an emotion meant to them. This approach was a deliberate decision in the design of the project as I wanted to the student's experiences to guide the research.

Future studies may also want to incorporate more open-ended questions that encourage participants to express their own reasoning in order to capture the nuanced experiences of students. Incorporating qualitative research methods might allow researchers to gain deeper insights into students' coping strategies and their interpretations of campus spaces.

5.5.3 Diversity of participants

Personal perspectives and internal experiences of students could impact how they perceive and process emotion or even recognize mental health and wellbeing. Various factors

like personal history, beliefs, values, and cultural background influence subjectivity (Siegesmund, 2008; Wilson & Sharples, 2015). Thus, when it comes to emotions and experiences in campus spaces, different students may have distinct subjective realities. The following paragraphs discuss specific limitations that may have contributed to the results of my study.

5.5.3.1 Year of study

There was a pre-requisite in the questionnaire stating that by agreeing to participate, students committed to having familiarity and experience with the campus built environment. The majority of students that participated in the questionnaire (61%) were in first or second year university. Given the gradual return to campus started in Fall 2021 following COVID 19 public health measures, by the time most of the students participated in the study, they would have had six-to-nine months of some form of exposure to the campus built environment. Although the public health measures eased over the course of the academic year, the restrictions or limitations on students' movement on campus remained in place, thus possibly influencing how they engaged with the campus built environment.

5.5.3.2 Gender

Another factor that may have impacted the results is the imbalance in gender self-identification in that the majority of the respondents were female (76%). For instance, research indicates that women tend to use more emotional coping and social support seeking strategies, while men may lean towards problem-solving approaches (Tamres et al., 2002; Meléndez et al., 2013). Another example of gender bias may be evident in the types of campus spaces shared by students. One study indicates that women exhibit greater active engagement, value aesthetics, and experience wellbeing associated with green spaces, suggesting a stronger connection to and appreciation for nature compared to men (Ode Sang et al., 2016). The number of respondents

who identified as female in my study may have impacted the self-reporting of spaces that were in nature or outdoors. Future studies might benefit from either a more balanced pool of respondents or probing any relationships that may exist between types of spaces (e.g., outdoors) and gender.

5.5.3.3 Ethnocultural groups

I asked students to share the perceived influence that design features that support diverse cultural practices may have on them, but I did not create a purposeful recruitment strategy to reach a culturally diverse student population. Additionally, I did not include culturally specific mechanisms (e.g., norms, values, beliefs, or practices) (Matsumoto & Hyisun, 2022) as part of the study. Without these mechanisms, there would be difficulty in understanding how culture influences or contributes to student mental health and wellbeing, particular as it relates to emotion as culture plays a role in shaping the processes of emotion (Chiao, et al., 2022). These inactions may have influenced the diversity of respondents (the majority of participants who answered the demographic questions identified as white/caucasian (56%), with the remaining participants identifying from different ethnocultural groups (e.g., Asian, Black, Middle Eastern)). Future studies may benefit from a purposeful strategy to recruit a more diverse participant group and to work with equity-deserving groups to better understand how such biases can be addressed in the study, including the design, dissemination, and analysis of the questionnaire.

5.5.3.4 Religion

I also asked students the level of consideration that they gave to religious-based coping strategies when self-selecting space on campus, I did not attempt to specifically recruit religious populations. Furthermore, I did not collect demographic information that would indicate what percentage of respondents may identify as religious or spiritual. These actions may result in a disproportionate number of students indicating that religion-based strategies were either of little consideration or not applicable.

5.5.4 Overall response rates

In part one of the questionnaire (where do students go), the response rate varied between 141 at the beginning of the section to 67 respondents at the end of the section. However, in part two of the questionnaire (where do students avoid), the number of respondents started at 60 and dropped to 47 students by the end of the questionnaire. The declining response rate may be due to participant fatigue given the survey length (64 questions that could take a minimum of 25 minutes to complete). This may have resulted in students exiting the questionnaire prematurely, or not being as thorough in their responses. As part of the questionnaire design, the anticipated length of time and progress bar were added to try and mitigate this issue (Sharples & Cobb, 2015a). For future research, I would consider exploring split questionnaire design, which organizes the questions into different modules and randomly assigns participants to a subset of these modules (Axenfeld et al., 2022). The phrasing and positioning of the *avoid* questions may have contributed to declining response rates as well. For example, I asked students to consider why they might avoid the space they named in the first part of the question. If they responded by giving strong consideration to hanging out with friends, it was not clear if the strong consideration was because the student wants to avoid the space to avoid hanging out with friends, wants to avoid the space because it does not support that coping strategy, or wants to avoid the space because their friends are not there, and they would prefer to see them. This could be mitigated in future studies by further testing of the questionnaire to determine if students interpret the questions in the same way (i.e., the reliability of the questions) (Sharples & Cobb, 2015b).

5.5.5 Response bias

I elected to repeat the pattern of questions that I posed to students in part one and two of the questionnaire. This decision could result in students developing a consistent pattern of responses throughout the questionnaire, either by selecting the same option repeatedly or by choosing a particular response style (e.g., always selecting extreme options) (*What Is Response Bias and How Can You Avoid It?*, n.d.). This can result in biased data and inaccurately represent the range of emotions and coping strategies individuals employ.

Social biases may also have impacted the meaning of the data. Respondents may have provided answers that they believe were socially desirable or acceptable rather than expressing their true response (Grimm, 2010) to emotion statements or consideration of coping strategies. Students may have felt pressured to give responses that align with societal expectations or that portray them in a positive light (Latkin, 2017).

5.6 Recommendations for future research

While my study demonstrates early findings regarding the campus built environment and student mental health and wellbeing, future research is needed to specifically examine the confluence of these topics. This is particularly critical as we see increasing rates of students grappling with mental health concerns including depression, overwhelm, and hopelessness (“New Canadian Standard Launched for the Mental Health of Postsecondary Students,” 2020; *The New Abnormal: Student Mental Health Two Years into COVID-19*, n.d.).

More specifically, one of the key findings in my research was that of potential patterns when students experienced perceived negative emotions. For instance, when students experienced anger, disgust, or sadness, they were more likely to give consideration to emotion- or avoidance-based coping strategies in their choices of where to go on campus. Based on this

finding, it may be interesting to adopt a salutogenic design mindset to explore what types of spaces and design features might support the processing of negative emotions or the employment of problem-based coping strategies as part of a student's sense of coherence. Based on this, future research could consider:

1. How do specific elements of the university built environment influence the experience and expression of negative emotions (e.g., sadness, anger, and disgust) among students?
2. What types of spaces and design features might promote the use of problem-based strategies among students?
3. Are there specific architectural interventions that can be implemented within the campus built environment to support the processing of negative emotions (e.g., sadness and disgust) including less well-understood or under-examined spaces (e.g., washrooms or parkades)?

In addition to the specific questions above, future research could delve deeper into the specific aspects of certain design features that students considered appealing or unappealing. One way to explore this may be to use spatial classifications, including fixed (e.g., structural elements), semifixed (e.g., furniture), and nonfixed features (e.g., behaviours) (Nasrallah & Pati, 2021). Moreover, researchers could explore design features by assessing the perceived importance and perceived effectiveness, both before and after any interventions or changes, to evaluate the impact of such modifications. This approach enables a more comprehensive understanding of the effects of design features on students' mental health and wellbeing.

Chapter 6: Conclusion

The aim of my research was to investigate the relationship between the built environment on post-secondary campuses and the mental health and wellbeing of students. Post-secondary institutions represented an interesting environment in which to conduct the research because of:

1. the mental health crisis among today's youth, particularly impacting the demographic pursuing post-secondary education (Linden et al., 2021; Mental Health - Ontario's Universities, n.d.; Mental Illness and Addiction: Facts and Statistics | CAMH, n.d.), and
2. the responsibility that post-secondary institutions have for the health, wellbeing, and safety of their students (Shanahan, 2019) .

6.1 Overall findings

My first step was to determine the extent of the gap in academic research regarding the built environment on post-secondary campuses and student mental health. In the initial phase of this study, I conducted a rapid scoping review to lay the foundation for a comprehensive investigation into the academic literature on the post-secondary campus built environment and its impact on the mental health and wellbeing of students. This review was instrumental in guiding the subsequent stages of the research and provided valuable insights into existing knowledge gaps and potential avenues for exploration.

The results of the review suggested a potential relationship between the post-secondary campus built environment and students' mental health and wellbeing. This relationship encompassed various aspects of the campus, including learning environments, dormitories, and outdoor landscapes. The review indicated that the built environment may influence students' emotional processing, mood, reaction to stress, and cognitive function. This part of my study

provides designers with a synthesis of information currently available related to promoting positive mental health outcomes for students by considering factors such as lighting, sound, temperature controls, housing quality, personal control, and social support. The results highlight that it may be important for designers to look more broadly at if and how direct and indirect influences may work together to enable or dismantle barriers for students to access general resistance or specific resistance resources, both of which may contribute to an overall sense of coherence and wellbeing.

After conducting a rapid scoping review, I found that the existing research on the subject matter lacked a focus on understanding the campus built environment from the perspective of students. Previous studies in the review seemed to approach campus space as a functional component (e.g., students go to a café for food or beverage, or the library to study). However, it is possible that students experience the campus in various ways that go beyond this functional approach, indicating that space has multiple uses and contexts (e.g., students go to a café to do homework, or go to the library to connect with peers). By asking students how they experience space, designers can create campus environments that support students' sense of coherence and promote positive mental health outcomes.

The results of my student questionnaire found that student centres and libraries were important spaces for students' engagement with the campus. These spaces offered resources, study areas, and opportunities for socialization and connection with peers. Furthermore, the findings indicated that environmental factors such as noise and overcrowding seemed to be detractors for students when selecting spaces on campus. This finding aligned with previous research that highlighted the negative impacts that noise has on mental health and performance.

Additionally, the results of the questionnaire indicated that avoidance-based and emotion-based coping strategies, tended to motivate students' space selection, particularly when experiencing negative emotions. By understanding how students' emotions and coping strategies impact their space selection, designers can create campus environments that better support students' mental health and wellbeing. For example, if designers know that students tend to avoid certain spaces when feeling negative emotions, they can work to create alternative spaces that are more conducive to positive emotions and coping strategies or even create spaces where it is ok to just experience negative emotions.

In the last part of the study, I asked students to consider the influence design features may have in the spaces they chose to go. Students rated layout, orderly and clean spaces, access to natural light, and noise control as the most influential design features in their selection of campus spaces based on their emotions. These features were associated with comfort, concentration, and overall satisfaction in the environment. The layout of spaces was particularly important and was consistently ranked as the most considered feature across different emotions. By considering these findings, universities can improve their campus environments particularly by focusing on the spaces students go to or avoid.

6.2 Final remarks

In summary, this research study examined the relationship between the built environment on post-secondary campuses and the mental health and wellbeing of students. Through a rapid scoping review and subsequent empirical investigation, I found that there is a relationship between the campus built environment and student mental health and wellbeing.

The study makes important contributions to the field by expanding the knowledge base on the relationship between the built environment and student mental health and wellbeing. It

addresses a gap in the literature by specifically examining the perceived influence of coping strategies and design features on students' space choices. Practical implications of the research include the development of a matrix of design features to inform future design considerations in post-secondary environments.

However, the study had limitations, such as the assumption of students' self-awareness regarding their emotional states and the complexity of analyzing multiple variables.

Recommendations for future research include conducting more comprehensive empirical studies to examine the confluence of the campus built environment and student mental health, exploring the influence of specific elements on the experience of negative emotions, and understanding the reasons behind the prevalence of certain coping strategies among students when selecting spaces on campus.

Overall, this research study provides valuable insights into the relationship between the campus built environment and student mental health and wellbeing. Further research in this area will contribute to the development of evidence-based resources and the improvement of post-secondary environments to better support the mental health and wellbeing of students.

Ultimately, I hope to bring awareness to the importance of prioritizing student mental health in the design and implementation of post-secondary campus built environments.

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Appendices

Appendix A – Example of thematic sorting for rapid scoping review

Types of environment studied	Built environment direct Influences (Evans, 2003)	Built environment indirect Influences (Evans, 2003)	Salutogenic resources (Mittlemark, 2017)
Learning environment <i>"Learning environment refers to the diverse physical locations, contexts, and cultures in which students learn."</i> https://www.edglossary.org/learning-environment/	- Housing - Crowding - Noise - Indoor air quality - Light - Thermal	- Personal control - Socially supportive relationships - Restoration from stress and fatigue	GRRs - Physical* - Bio-chemical* - Cognitive* - Emotional* - Attitudinal* - Sociocultural* - Interpersonal*
Bowrey, David J., Kidd, Jane M.	Foo J., Mavrogianni A.	Bowrey, David J., Kidd, Jane M.	Bowrey, David J., Kidd, Jane M.
Foo J., Mavrogianni A.	Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.	Peters T., D'Penna K.	Peters T., D'Penna K.
Marchand, Gwen C.; Nardi, Nicholas M.; Reynolds, Douglas; Pamoukov, Stoil	Marchand, Gwen C.; Nardi, Nicholas M.; Reynolds, Douglas; Pamoukov, Stoil	Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.	Reese R.F., Seitz C.M., Gosling M., Craig H.
Licite L., Janmere L.	Yildirim K., Capanoglu A., Cagatay K.	Reese R.F., Seitz C.M., Gosling M., Craig H.	Marchand, Gwen C.; Nardi, Nicholas M.; Reynolds, Douglas; Pamoukov, Stoil
Yildirim K., Capanoglu A., Cagatay K.	Past experiences shape people's expectations of their thermal environment...affect response to thermal conditions (Willey 1987)	Licite L., Janmere L.	Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.
Peters T., D'Penna K.	environmental setting has a direct impact on perception, comfort, motivation, and concentration	achievements depend on the curricula, teaching techniques and social relations. Also on the physical environment of an educational institution...impact on the course of personality socialisation and culturalization as well as on its intensity	
Living environments			SRRs - Service - Infrastructure* - Facility* - Amenity* - Program
Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.	pertinent themes in published multi-disciplinary literature relating to the influence of the built environment on university students: emotional stress, happiness, stimulation, cognitive function, social support, belonging, places to study, lighting, and ventilation		Peters T., D'Penna K.
Outdoor space			Foo J., Mavrogianni A.
Reese R.F., Seitz C.M., Gosling M., Craig H.	Design elements impact perception of engineering controls	Physical environ impacts studying process	Marchand, Gwen C.; Nardi, Nicholas M.; Reynolds, Douglas; Pamoukov, Stoil
			Licite L., Janmere L.
			Yildirim K., Capanoglu A., Cagatay K.
			Cedeno Laurent JG, Allen JG, McNeely E, Dominici F, Spengler JD.

Appendix B – Questionnaire

Coping and the campus built environment

Survey Flow

Standard: Introduction (2 Questions)
Standard: Research Ethics (2 Questions)
Standard: Official consent (2 Questions)

Branch: New Branch

If

If I have read and understood what the survey is about as well as any risks associated with it. I co... No, I will not participate Is Selected

Standard: Survey Withdrawal (1 Question)

EndSurvey: Advanced

Standard: Where you go and why (25 Questions)
Standard: Where you avoid and why (25 Questions)
Standard: Demographics (3 Questions)
Standard: Equity information (4 Questions)
Standard: Debriefing (1 Question)
Standard: Lottery and Future research (1 Question)

EndSurvey: Advanced

Page Break

Start of Block: Introduction

Summary

This survey explores what spaces you use on campus based on how you feel.

Background

Research demonstrates that human-made environments affect mental wellness which can contribute to or deter someone's wellbeing, motivation, engagement, belongingness. This research project contributes to a larger area of inquiry by exploring if students use the human-made environment as part of their overall coping strategies. The project will accomplish this by exploring student experiences and the use of campus spaces.

Survey Description

The following survey lists a series of six (6) emotions that we all experience at different times. You will be asked to identify what places you go on campus when you feel that way and why. You will also be asked what spaces you might avoid on campus when you feel that way and why.

You may find it helpful to keep a copy of the campus map open as a reference although an embedded link to the map is provided in the relevant questions.

The survey is hosted on Qualtrics. The questions are detailed so you may find it easier to participate from a computer rather than a mobile device.

Please complete it **by April 8, 2022**, to be included in the lottery.

Expected Duration

20-30 minutes

Page Break

Eligibility

The following conditions are required to participate in the survey:

- **18 years of age or older**
- **Ability to read and write in English**
- **Current undergraduate or graduate student at Carleton with a valid Carleton email address**
- **Familiarity and experience with Carleton's physical campus**

Compensation

There is no compensation for participating in this research.

Chance to win!

In appreciation of their time, all Carleton student participants who complete the survey may enter a **lottery to win a \$50** top-up to their campus card (or a gift card of the same value to either Amazon.ca or Starbucks). There will be one (1) name selected for every 25 participants in the lottery until the **survey closes on April 8, 2022**.

At the end of the survey, you will be directed to a separate survey form to enter the lottery. The draw will be on **April 9, 2022**. Only participants with a valid Carleton email address will be eligible.

Questions about the survey or research?

For questions about the study, please contact cathyedwards@cmail.carleton.ca.

If you have any ethical concerns with the study, please contact the Carleton University Research Ethics Board-B by email at ethics@carleton.ca.

End of Block: Introduction

Start of Block: Research Ethics

How is your data secured?

No IP addresses will be collected by Qualtrics although a cookie will be placed in the browser used to complete the survey so that it is only taken once. Survey responses will be stored on and protected by Qualtrics, in a server located in Canada and could be disclosed in the event of a court order or data breach.

Survey responses are anonymous. They will be downloaded and stored on Citrix for the purpose of research. This service is hosted at Carleton University and all of the research data is securely stored in Carleton's data centres. Only the research team has an account

in this folder and will have access to the data. The data is protected by password and two-factor authentication for all users. Responses will be kept for seven (7) years for use in possible publications, presentations, and future research.

A spreadsheet containing names and contact information of participants who participate in the draw will also be kept on Citrix for no longer than three (3) months in order to administer the lottery.

Page Break

Consent to participate

By selecting the "Yes, I will participate" option, you agree to participate in the survey. Any associated risks with taking the survey are expected to be low. Some questions may create emotional discomfort or disturbance for some participants. Please know that you can skip any question, take breaks, or come and go from the survey without loss of previous answers.

Withdrawal from the study

Participants can withdraw consent at any time during the survey process. As every effort has been made to anonymize the data, it is not possible to withdraw from the study once a participant submits their survey.

Ethics approval

This study has been reviewed and received **Research Ethics Clearance #117115** through the Carleton University Research Ethics Board-B.

End of Block: Research Ethics

Start of Block: Official consent

By click on "I confirm that I am eligible" you are confirming that you are:

1. 18 years of age or older
2. Able to read and write in English
3. Current undergraduate or graduate student at Carleton with a valid Carleton email address
4. Familiar with Carleton's campus

☐

I confirm that I am eligible (4)

☐

I am not eligible (5)

I have read and understood what the survey is about as well as any risks associated with it. I consent to voluntarily participate in the study.

☐ Yes, I will participate (1)

☐ No, I will not participate (2)

End of Block: Official consent

Start of Block: Survey Withdrawal

Survey Withdrawal

Thank you for taking the time to consider the invitation to participate in the survey. Any data captured will be deleted.

If you wish to elaborate on the reasons why you chose NOT to participate in the survey, you can provide details in the box below.

End of Block: Survey Withdrawal

Start of Block: Where you go and why

Part 1: Where do you go? The first series of questions focus on **where you go on campus** when you feel certain emotions (joy, sadness, fear, anger, surprise, or disgust). If you are not sure what a particular emotion means, please use your best judgement. There are no right or wrong answers. The survey is open to interpretation and meant to capture your experiences and impressions.

Page Break

Short scenario: You are feeling **happy or joyous**.

Where do you go on campus? Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

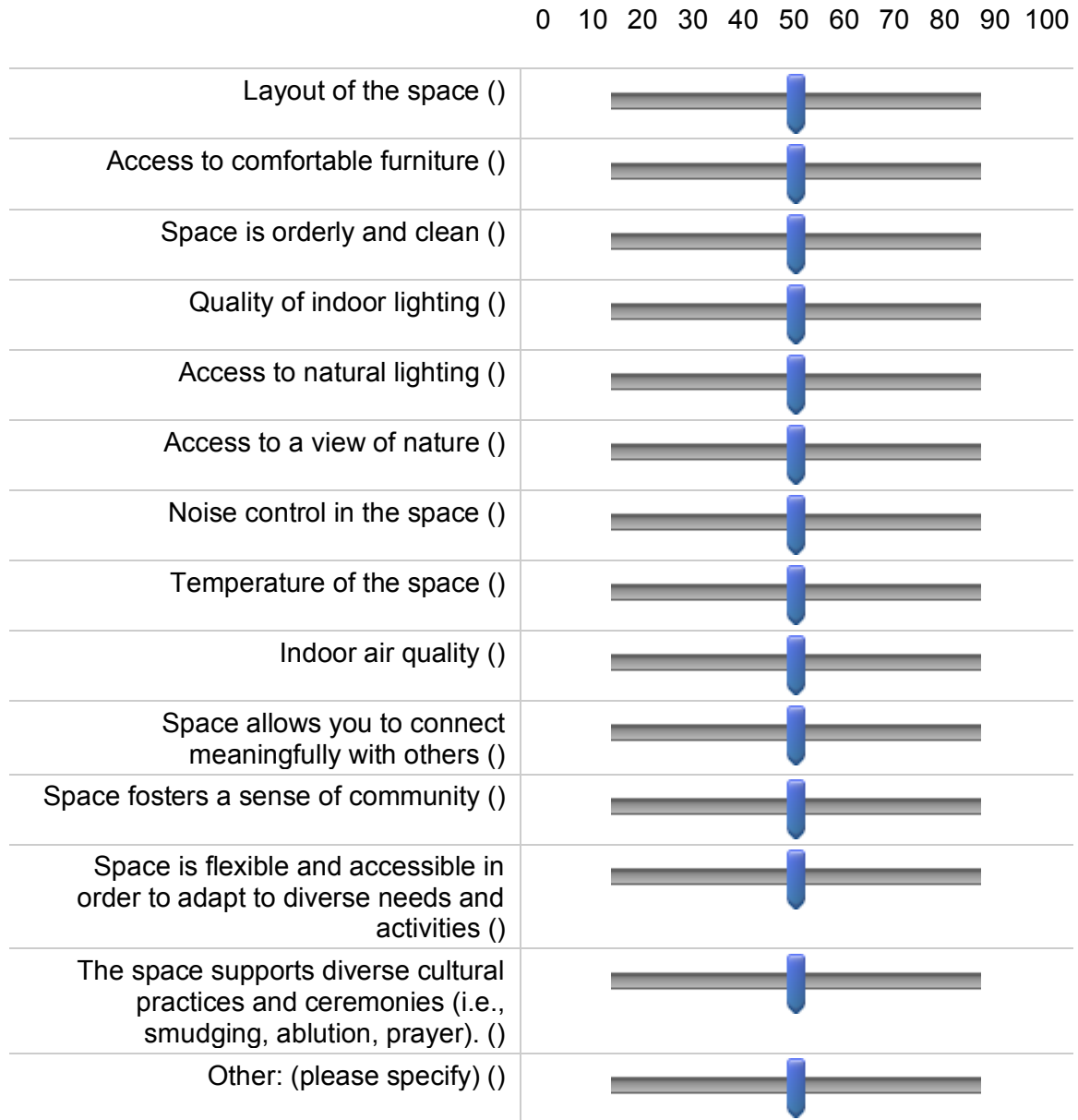
How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there. (0 means the listed feature has no

influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are feeling **sad or down**.

Where do you go on campus? Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

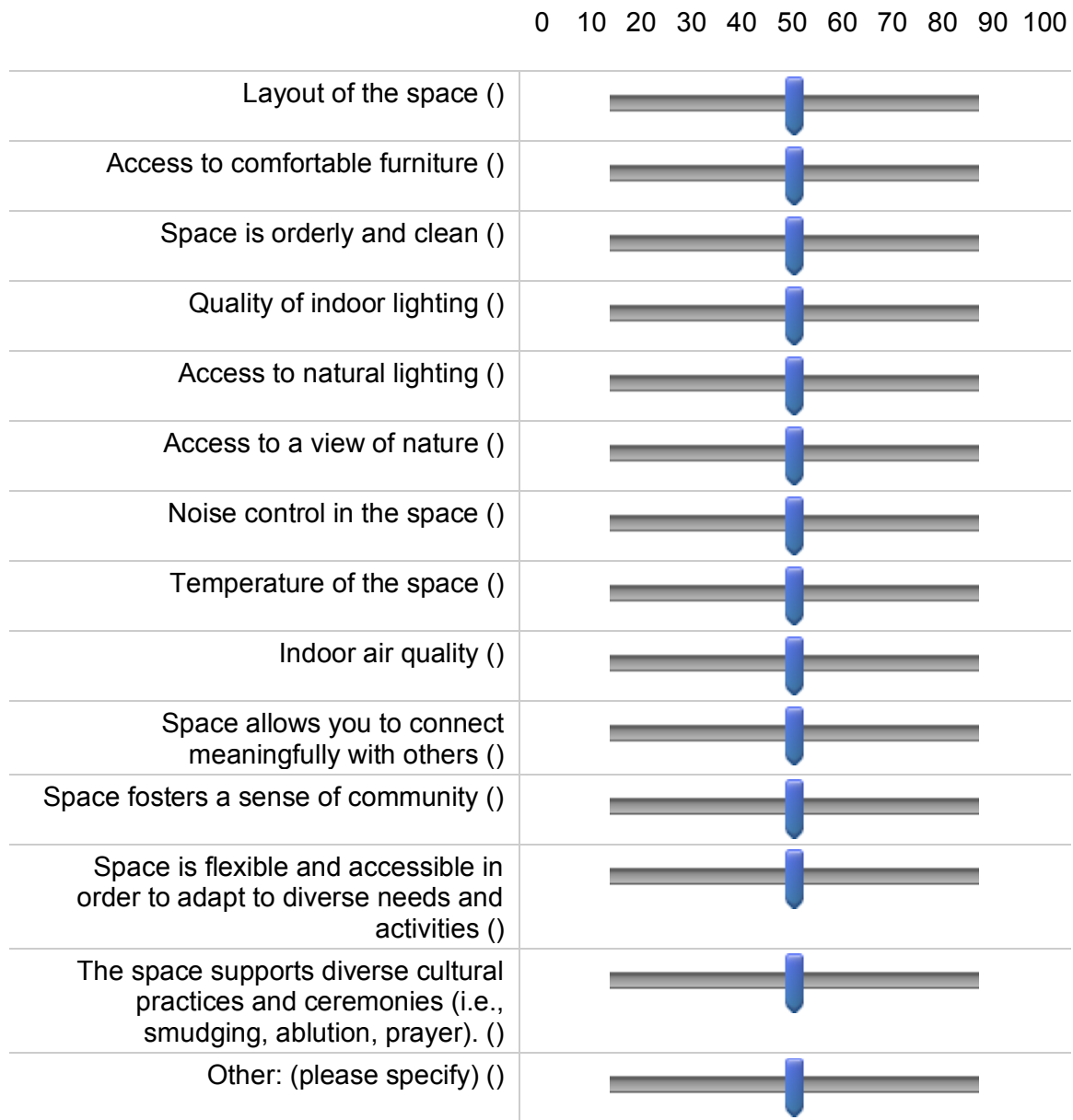
How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are feeling **fearful or afraid**.

Where do you go on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

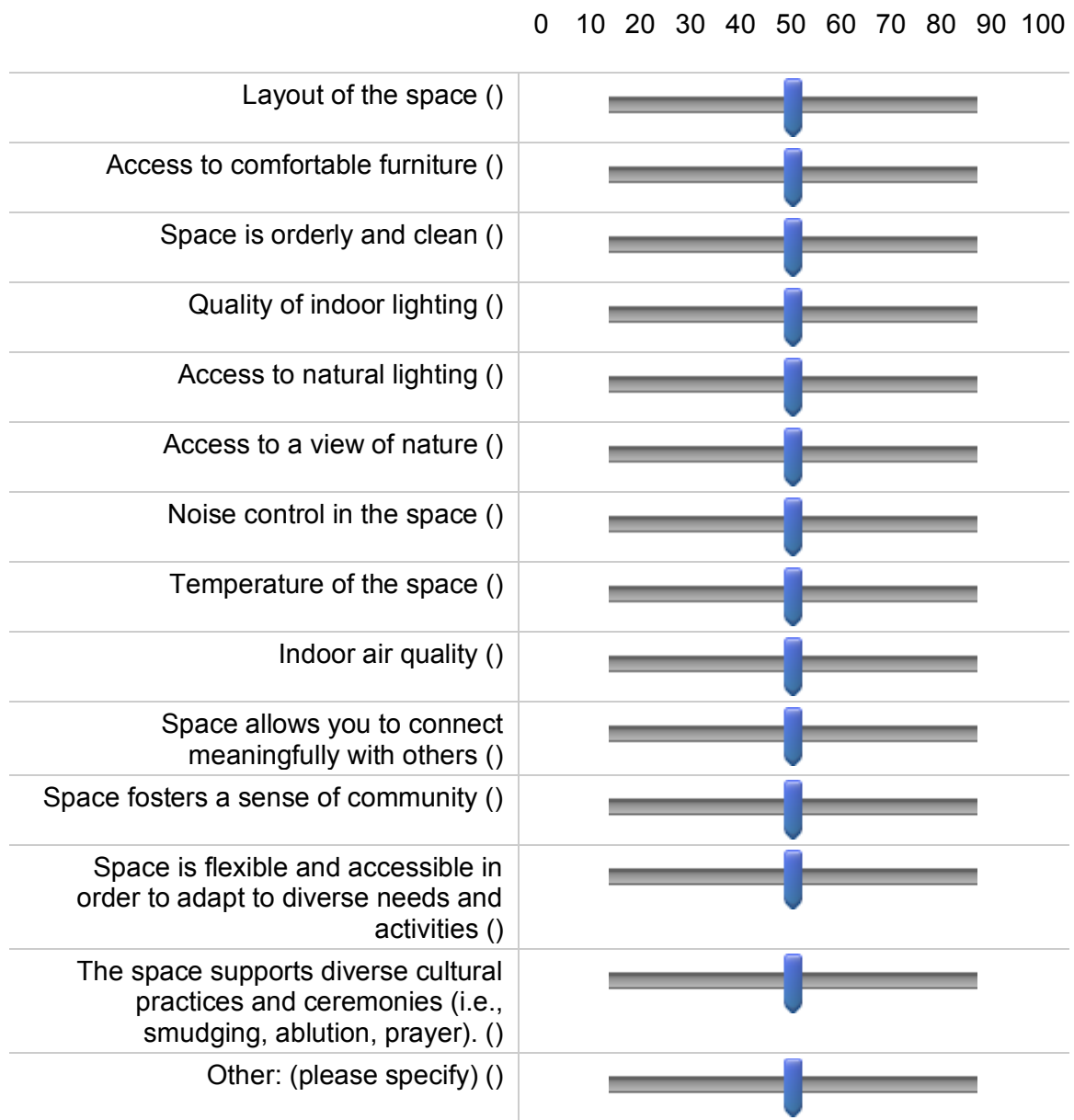
How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are **frustrated or angry**.

Where do you go on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map

below.

[Map](#)

Page Break

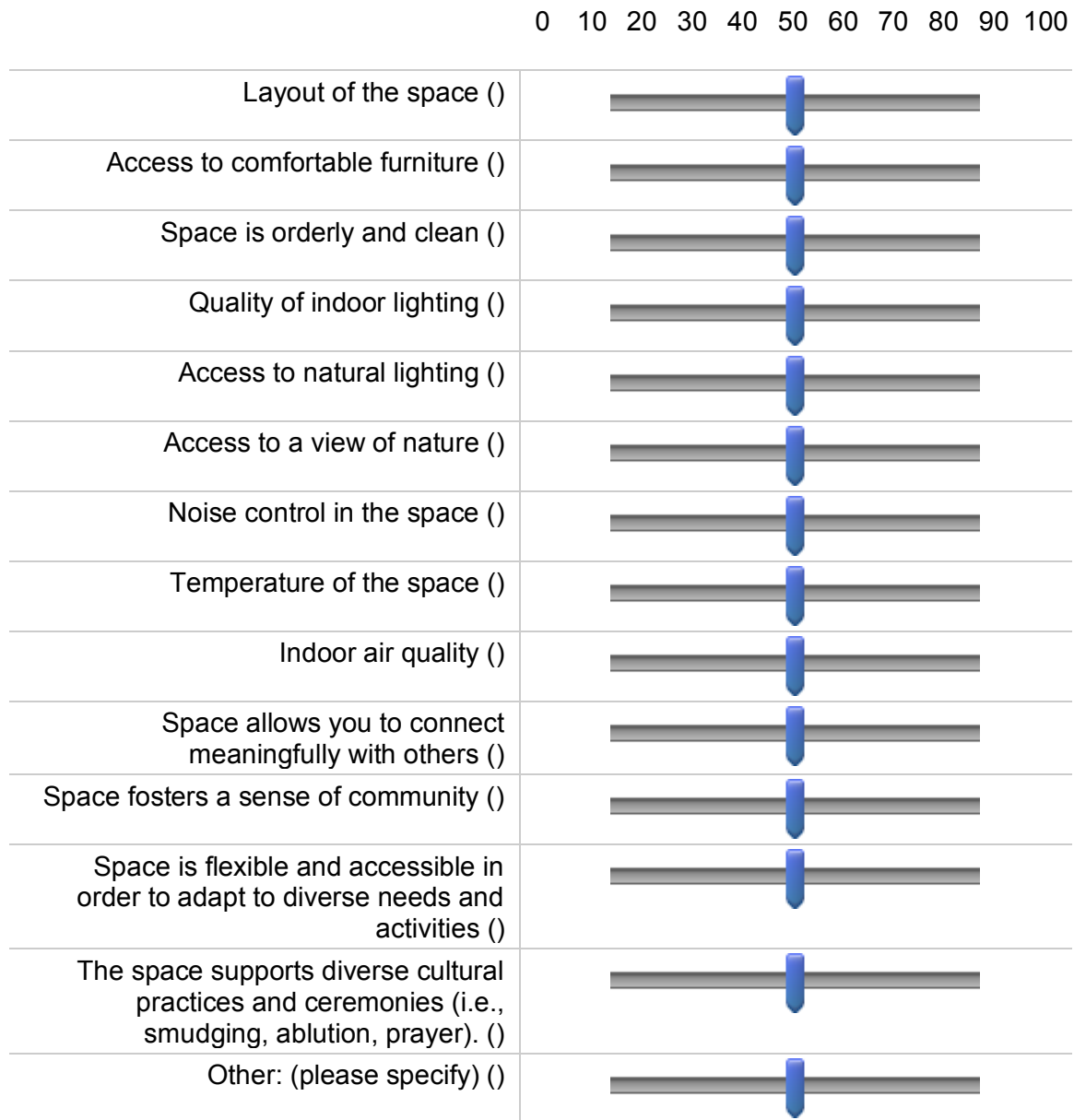
How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are **surprised** by something.
Where do you go on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).

0 10 20 30 40 50 60 70 80 90 100

Layout of the space ()	
Access to comfortable furniture ()	
Space is orderly and clean ()	
Quality of indoor lighting ()	
Access to natural lighting ()	
Access to a view of nature ()	
Noise control in the space ()	
Temperature of the space ()	
Indoor air quality ()	
Space allows you to connect meaningfully with others ()	
Space fosters a sense of community ()	
Space is flexible and accessible in order to adapt to diverse needs and activities ()	
The space supports diverse cultural practices and ceremonies (i.e., smudging, ablution, prayer). ()	
Other: (please specify) ()	

Page Break

Short scenario: You are feeling **disgusted or offended** by something or someone that you encountered.

Where do you go on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

How strongly do the following considerations factor into your decision of where to go on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What about this particular space appeals to you?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to go there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).

0 10 20 30 40 50 60 70 80 90 100

Layout of the space ()	
Access to comfortable furniture ()	
Space is orderly and clean ()	
Quality of indoor lighting ()	
Access to natural lighting ()	
Access to a view of nature ()	
Noise control in the space ()	
Temperature of the space ()	
Indoor air quality ()	
Space allows you to connect meaningfully with others ()	
Space fosters a sense of community ()	
Space is flexible and accessible in order to adapt to diverse needs and activities ()	
The space supports diverse cultural practices and ceremonies (i.e., smudging, ablution, prayer). ()	
Other: (please specify) ()	

End of Block: Where you go and why

Start of Block: Where you avoid and why

Part 2: Where do you avoid?

The second series of questions focus on **where you avoid on campus** when you feel certain emotions (joy, sadness, fear, anger, surprise, or disgust). If you are not sure what a particular emotion means, please just use your best judgement. There are no right or wrong answers. The survey is open to interpretation and meant to capture your experiences and impressions.

Page Break

Short scenario: You are feeling **happy or joyous**.

Where do you avoid going on campus? Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

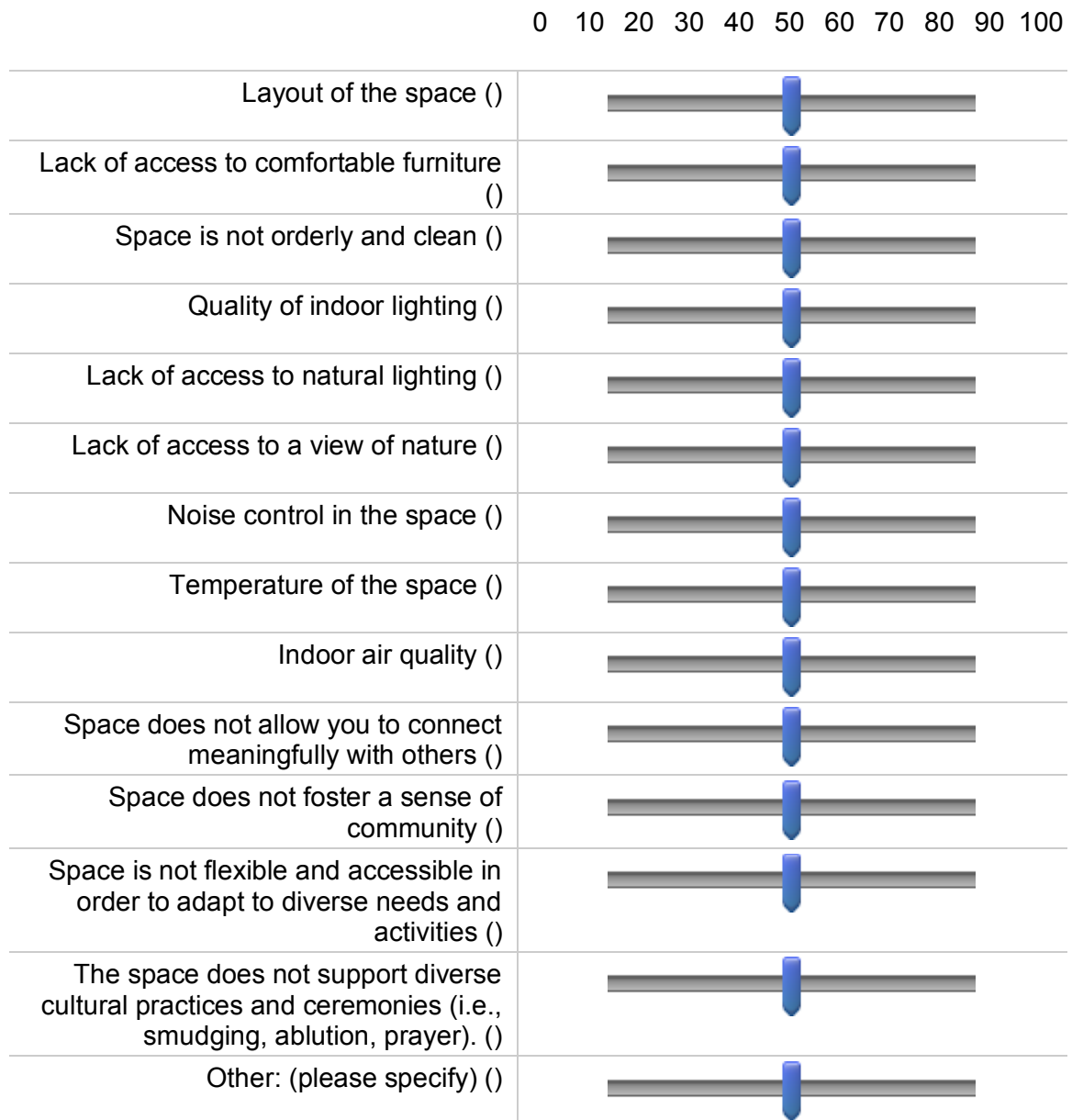
How strongly do the following considerations factor into your decision to avoid this space on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are feeling **fearful or afraid**.

Where do you avoid going on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

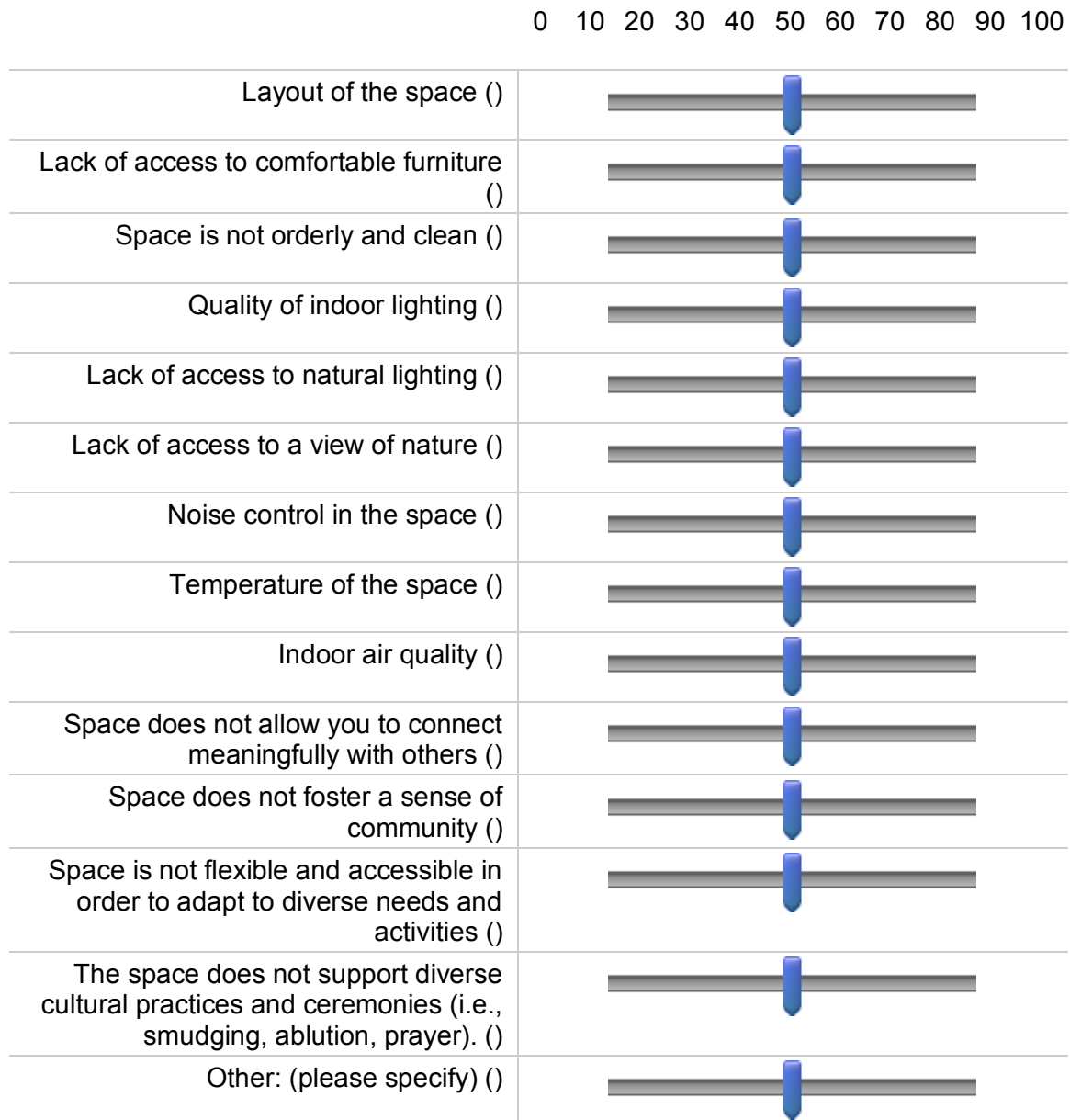
How strongly do the following considerations factor into your decision to avoid this space on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



 Page Break

Short scenario: You are feeling **sad or down**.

Where do you avoid going on campus?

Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference

point by clicking on the word map below.

[Map](#)

How strongly do the following considerations factor into your decision to avoid this space on campus?

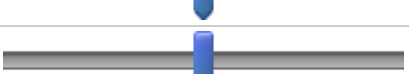
	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).

0 10 20 30 40 50 60 70 80 90 100

Layout of the space ()	
Lack of access to comfortable furniture ()	
Space is not orderly and clean ()	
Quality of indoor lighting ()	
Lack of access to natural lighting ()	
Lack of access to a view of nature ()	
Noise control in the space ()	
Temperature of the space ()	
Indoor air quality ()	
Space does not allow you to connect meaningfully with others ()	
Space does not foster a sense of community ()	
Space is not flexible and accessible in order to adapt to diverse needs and activities ()	
The space does not support diverse cultural practices and ceremonies (i.e., smudging, ablution, prayer). ()	
Other: (please specify) ()	

Page Break

Short scenario: You are **frustrated or angry**.

Where do you avoid going on campus? Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

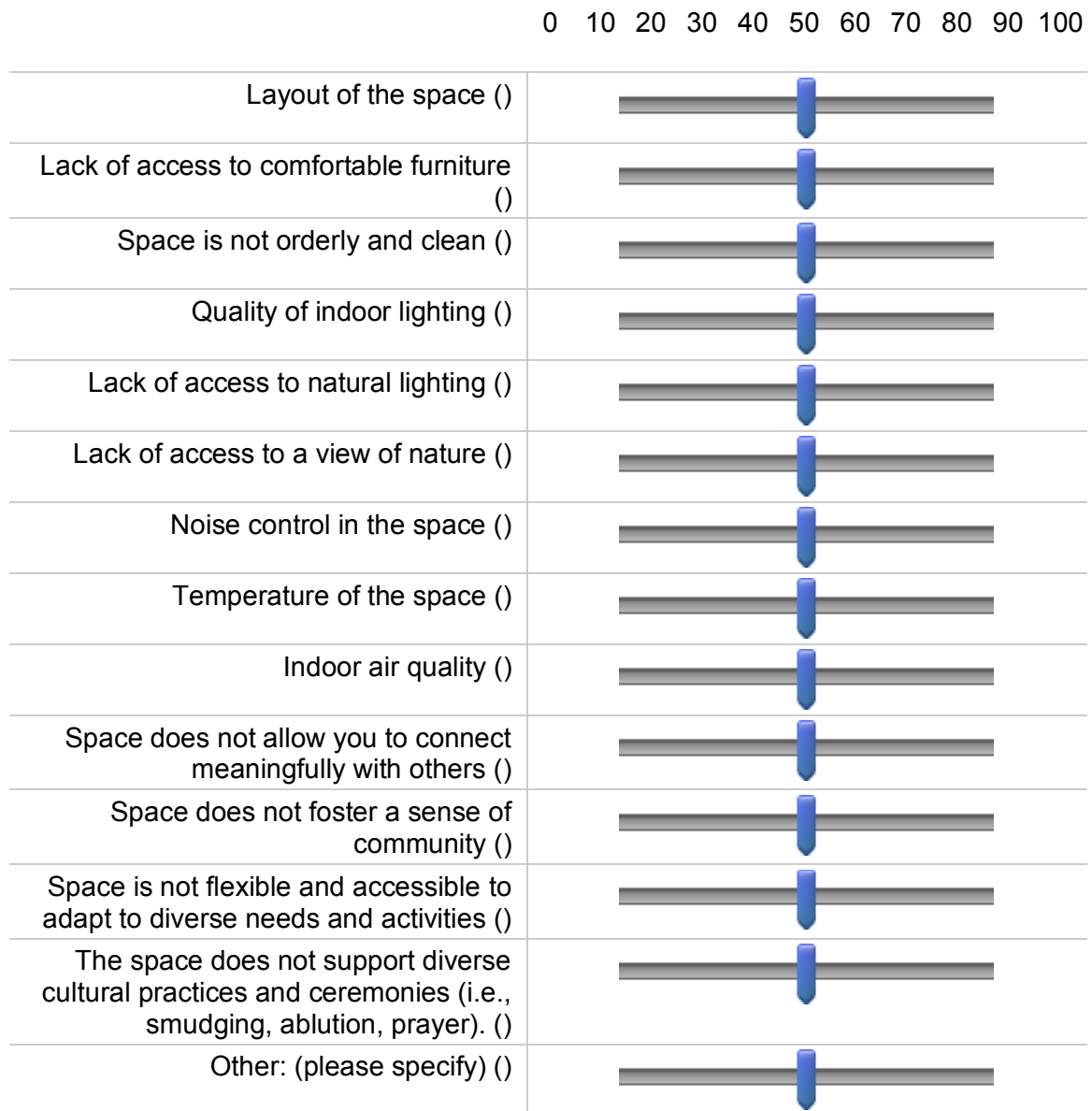
How strongly do the following considerations factor into your decision to avoid this space on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Page Break

Short scenario: You are **surprised** by something.

Where do you avoid going on campus? Please name one specific space on campus in the box below.

Try to be as specific as possible. You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

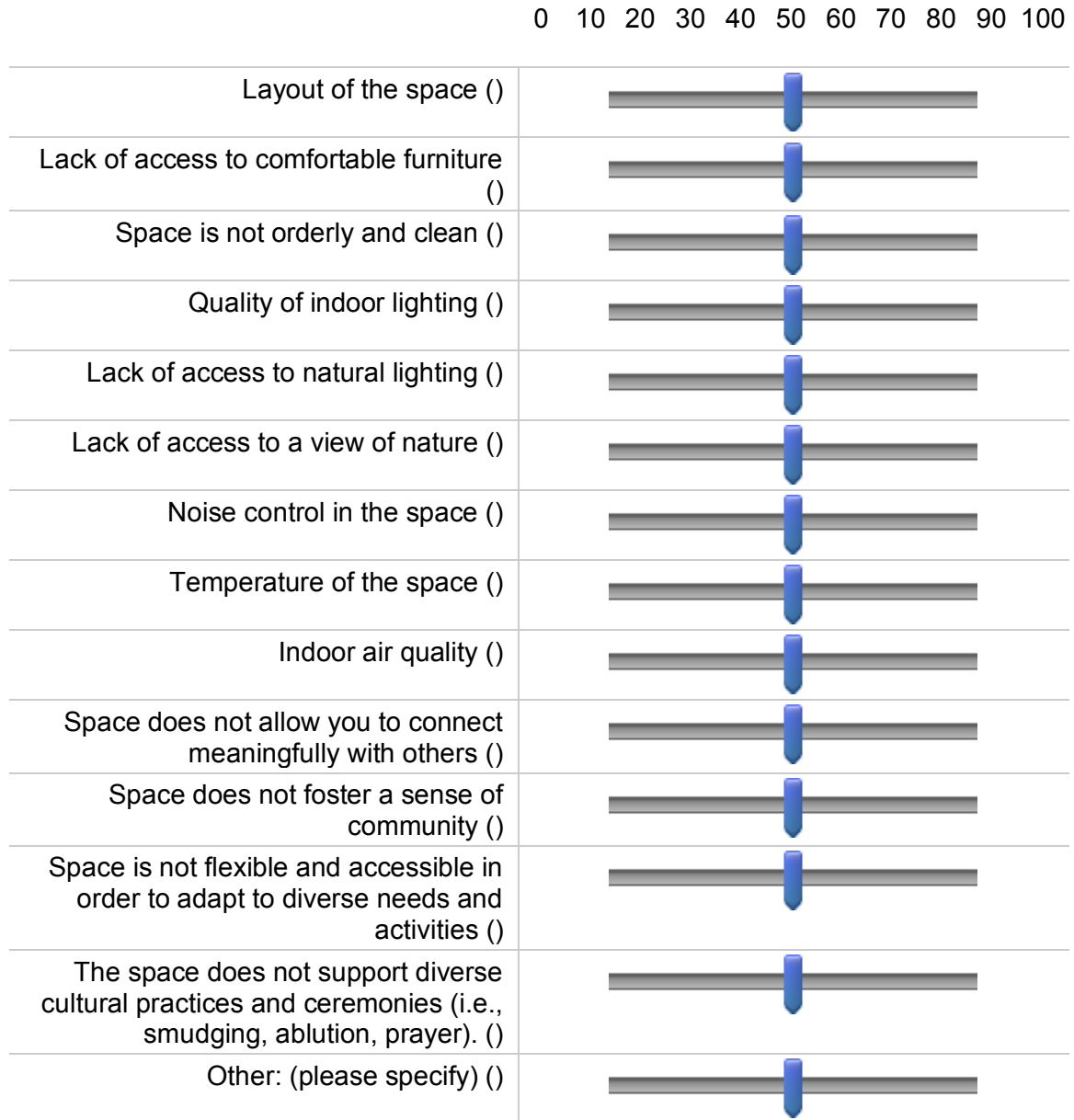
How strongly do the following considerations factor into your decision to avoid this space on campus?

	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



Short scenario: You are feeling **disgusted or offended** by something or someone that you encountered.

Where do you avoid going on campus?

Please name one specific space on campus in the box below. Try to be as specific as possible.

You can refer to the campus map to act as a reference point by clicking on the word map below.

[Map](#)

Page Break

How strongly do the following considerations factor into your decision to avoid this space on campus?

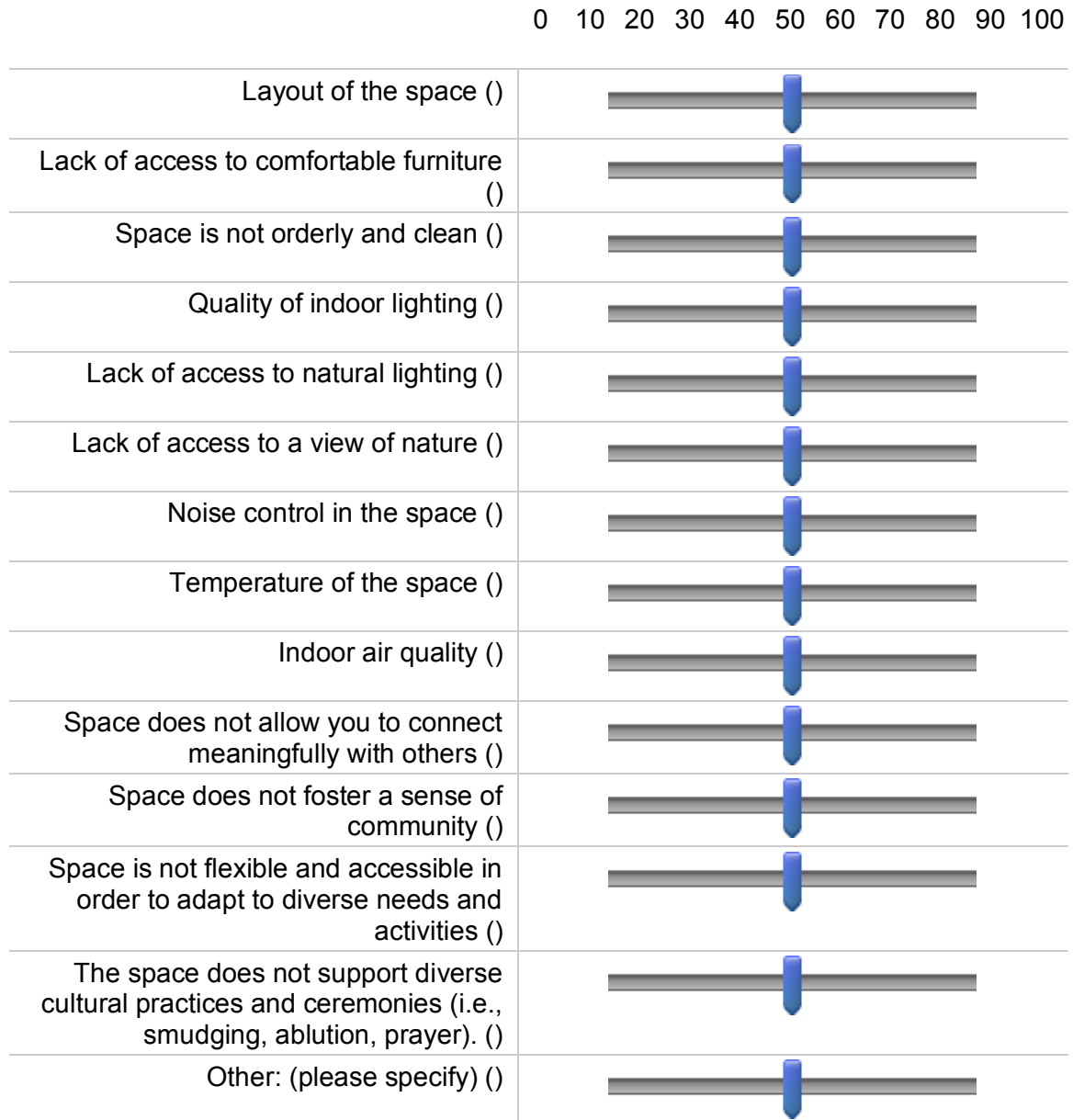
	Little consideration (1)	Moderate consideration (2)	Strong consideration (3)	Not applicable (4)
Hang out with friends (1)	☐	☐	☐	☐
Release emotion or 'let off steam' (2)	☐	☐	☐	☐
Avoid source or cause of emotion (3)	☐	☐	☐	☐
Access resources to support you (4)	☐	☐	☐	☐
Spiritual needs (5)	☐	☐	☐	☐
Other: (please specify) (6)	☐	☐	☐	☐

What is unappealing about this space?

Using the scale below, please adjust the slider to reflect how strongly a particular feature

in this space influences your decision to avoid going there.

(0 means the listed feature has no influence and 100 means the feature has the strongest influence).



End of Block: Where you avoid and why

Start of Block: Demographics

What age are you?

- ☐ 18-23 years old (2)
- ☐ 24-29 years old (3)
- ☐ 30-35 years old (4)
- ☐ 36-41 years old (5)
- ☐ 42 years old or more (6)

What is your current level of study?

- ☐ Undergraduate (1)
- ☐ Masters (2)
- ☐ PhD (3)
- ☐ Post-doctoral (4)
- ☐ Other (5) _____

What year are you currently in for your studies

- ☐ First year (4)
- ☐ Second year (5)
- ☐ Third year (6)
- ☐ Fourth year (7)
- ☐ Fifth year (8)
- ☐ Sixth year or more (9)

End of Block: Demographics

Start of Block: Equity information

Are you the first generation in your family attending university- or college-level studies?

- ☐ Yes (1)
- ☐ No (2)
- ☐ Unsure (4)

Do you identify as having a disability or as disabled (including visible and non-visible characteristics)

- ☐ Yes (1)
- ☐ No (2)
- ☐ Other (3) _____

What ethnic group(s) do you identify as? Please list all relevant groups in your own words.

What gender(s) do you identify as? Please list all relevant genders in your own words.

End of Block: Equity information

Start of Block: Debriefing

Thank you for participating in this survey. We are grateful for your time and reflections. The next page contains information regarding the lottery for a gift card.

If any questions have caused you discomfort or disturbance, please reach out for help and support.

Carleton Health & Counselling

Phone: 613-520-6674

Website: <https://carleton.ca/health/counselling-services/>

Carleton Spirituality Centre

Connect with one of the spirituality care team members by accessing their profiles.

Website: <https://carleton.ca/wellness/spirituality/>

Off-campus resources

Good2Talk provides free, confidential support services for post-secondary students.

Website: <https://good2talk.ca/>

Counselling Connect provides quick access to a free phone or video counselling session.

Website: <https://www.counsellingconnect.org/>

End of Block: Debriefing

Start of Block: Lottery and Future research

You have reached the end of survey. Thank you for participating.

Clicking on the next arrow will open the survey form for a chance to win a **\$50** top-up to your campus card (or a gift card of the same value to either Amazon.ca or Starbucks).

There will be one (1) name selected for every 25 participants in the lottery until the **survey closes on April 8 2022**.

Please be sure to select the appropriate response in the form indicating your wish to participate in the lottery and/or future research. Note that contact information is not associated with survey responses, thus keeping your responses and data anonymous.

Appendix C – Campus map of the university

